

Coupled Training with Privileged Features and Unlabeled Data

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Abstract

In many prediction problems, we have extra information during training (for example, measurements that are expensive or slow to collect) that will not be available when the model is deployed. A common strategy is to first train a model that uses all training information, then use its predictions on unlabeled examples to train a second model that only uses the inputs available at test time. However, when the extra training-only information is weak or noisy, this two-stage approach can mislead the deployment model and even hurt accuracy. We propose a joint training method that learns the two models together, so the deployment model can benefit from the extra information only when it actually helps, instead of inheriting its mistakes. We provide guarantees that describe when joint training improves prediction accuracy and analyze a simple alternating training algorithm for large, high-dimensional models. Experiments on synthetic data and medical prediction tasks show that our approach avoids these failures and consistently outperforms standard two-stage baselines.

1. Introduction

Many modern learning settings provide extra information during training that cannot be used at deployment. Alongside the primary features X (e.g., an image or routine clinical measurements) and responses Y (e.g., a continuous outcome such as disease severity or a lab value), training examples may also include privileged features W (e.g., expensive biomarkers, specialist assessments). Since W can be unavailable, costly, or impractical to obtain at test time, the deployed predictor must depend only on X . At the same time, a large fraction of the training data may contain W but not the true labels. The goal is therefore to use W to improve training while still producing a final predictor that uses only X . This

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setting is motivated by many applications, including medical domains where reliable labels are expensive (Rajkomar et al., 2019; Fries et al., 2019), forecasting and longitudinal studies where intermediate or future measurements are observed only during training (Hogan et al., 2004), as well as transfer learning (Zhuang et al., 2020) and distribution shift settings (Koh et al., 2021).

Problem Formulation. We formalize this setting as follows. Let $\mathcal{Z} = \mathcal{X} \times \mathcal{W}$ and define $Z = (X, W) \in \mathcal{Z}$. We observe n labeled samples

$$\mathcal{D}_L = \{(Z_i, Y_i)\}_{i=1}^n \stackrel{\text{i.i.d.}}{\sim} P_{X,W,Y},$$

and $m := N - n \gg n$ unlabeled samples

$$\mathcal{D}_U = \{Z_j\}_{j=n+1}^N \stackrel{\text{i.i.d.}}{\sim} P_{X,W},$$

where $P_{X,W,Y}$ is a distribution on $\mathcal{X} \times \mathcal{W} \times \mathbb{R}$ and $P_{X,W}$ is its marginal over $\mathcal{Z}$.

Our goal is to estimate the regression function

$$\mu(x) := \mathbb{E}[Y \mid X = x],$$

which uniquely minimizes the squared risk $\mathbb{E}[(Y - f(X))^2]$ over measurable functions $f : \mathcal{X} \rightarrow \mathbb{R}$, assuming $\mathbb{E}[Y^2] < \infty$.

Given this setup, the central question is whether we can use the privileged information W to achieve better performance than training on X alone, while still ensuring that the deployed predictor depends only on X .

Two-Stage Procedure. One way to leverage privileged data is through a two-stage procedure. In the first stage, we estimate a pseudo-response function $\hat{g}$, which we refer to as the *rich-view model*, using all available labeled samples $\{(Z_i, Y_i)\}_{i=1}^n$ by minimizing the empirical risk

$$\hat{g} \in \operatorname{argmin}_{g \in \mathcal{G}} \frac{1}{n} \sum_{i=1}^n (Y_i - g(Z_i))^2.$$

The fitted function $\hat{g}$ is then used to transfer knowledge by imputing pseudo-responses $\hat{Y}_j = \hat{g}(Z_j)$ for the unlabeled samples $\{Z_j\}_{j=n+1}^N$. In the second stage, we estimate the target function $\hat{f}$, which we will refer to as the *deployment model*,

from the combined dataset $\{(X_i, Y_i)\}_{i=1}^n \cup \{(X_j, \hat{Y}_j)\}_{j=n+1}^N$ by minimizing

$$\hat{f} \in \operatorname{argmin}_{f \in \mathcal{F}} \frac{1}{N} \left(\sum_{i=1}^n (Y_i - f(X_i))^2 + \sum_{j=n+1}^N (\hat{Y}_j - f(X_j))^2 \right).$$

This general strategy was studied in Hou et al. (2023) for specific parametric models and extended to more general nonparametric settings in Xia & Wainwright (2024).

Although it is appealingly simple, this two-stage pseudo-labeling strategy can be fragile when the privileged features W provide only weak or noisy information about the response. In such regimes, the first-stage estimator $\hat{g}$ may produce pseudo-responses that are poorly aligned with the true regression function μ , and these errors are then propagated to the second stage through the imputed labels. As a consequence, the deployment model $\hat{f}$ may be biased toward artifacts introduced by $\hat{g}$, leading to degraded performance relative to training on the labeled data alone. This phenomenon, often referred to as *negative transfer*, is especially pronounced when the predictive advantage of (X, W) over X is marginal, or when the privileged variables contain high-dimensional nuisance components that overwhelm their useful signal. In such settings, naive two-stage procedures can perform strictly worse than supervised learning using only the labeled samples, despite access to abundant unlabeled data (Xia & Wainwright, 2024). These limitations motivate approaches that more carefully control the influence of privileged information and adaptively balance the contribution of W during training.

Our Contributions. We summarize our main contributions as follows:

- **Coupled Training Framework:** We propose a unified estimation framework that jointly learns the deployment and rich-view models. By adaptively calibrating the influence of privileged information, our approach effectively mitigates the negative transfer often observed in two-stage pseudo-labeling methods.
- **Efficient Algorithms:** We develop an alternating minimization algorithm for the coupled objective and extend the framework to high-dimensional settings using a greedy forward selection procedure with provable optimization guarantees.
- **Theoretical and Empirical Analysis:** We provide theoretical insights into the conditions under which unlabeled data improves estimation and demonstrate, through synthetic and medical datasets, that our method consistently outperforms standard baselines when privileged signals are noisy or weak.

1.1. Pseudo-Response Calibration with Coupled Training

The main limitation of the two-stage approach is that the pseudo-responses produced by the rich-view model are treated as fixed targets in the second stage, with no mechanism to correct or attenuate their influence when the privileged signal is weak or noisy. To address this issue, we propose a coupled training procedure that jointly estimates the deployment model f and the rich-view model g through an alternating scheme. The core idea is to allow information to flow in both directions: while g provides pseudo-responses that enrich the effective sample size for learning f , the current estimate of f is in turn used to recalibrate g on the unlabeled data, subject to an explicit constraint that limits deviation from the labeled responses. This coupling yields an adaptive regularization of the privileged information and provides a principled mechanism for avoiding negative transfer.

We now describe the resulting algorithm and its associated optimization formulation.

Coupled Training Algorithm. Initialize with any feasible g_0 and fix a constraint level $\nu \geq 0$. For $k = 1, 2, \dots$, alternate until convergence:

1. **Update f .** Given the current rich-view model $\hat{Y} = g_{k-1}$, update f by solving

$$f_k \in \operatorname{argmin}_{f \in \mathcal{F}} \frac{1}{N} \left(\sum_{i=1}^n (Y_i - f(X_i))^2 + \sum_{j=n+1}^N (g_{k-1}(Z_j) - f(X_j))^2 \right).$$

2. **Recalibrate g .** Given the current deployment model f_k , update g by solving

$$\begin{aligned} g_k &\in \operatorname{argmin}_{g \in \mathcal{G}} \frac{1}{m} \sum_{j=n+1}^N (g(Z_j) - f_k(X_j))^2 \\ \text{s.t.} \quad &\frac{1}{n} \sum_{i=1}^n (Y_i - g(Z_i))^2 \leq \nu. \end{aligned}$$

If $\mathcal{F} \subset L^2(X)$ and $\mathcal{G} \subset L^2(Z)$ are convex function classes, and the constraint set is feasible, then the coupled alternating updates constitute an exact two-block coordinate minimization method for the joint convex optimization problem

$$\begin{aligned} \min_{(f,g) \in \mathcal{F} \times \mathcal{G}} \quad &\frac{1}{N} \left(\sum_{i=1}^n (Y_i - f(X_i))^2 + \sum_{j=n+1}^N (g(Z_j) - f(X_j))^2 \right) \\ \text{s.t.} \quad &\frac{1}{n} \sum_{i=1}^n (Y_i - g(Z_i))^2 \leq \nu. \end{aligned} \quad (1)$$

In particular, the objective decreases monotonically and any limit point of (f_k, g_k) is a global solution of (1) (Grippe & Sciandrone, 2000).

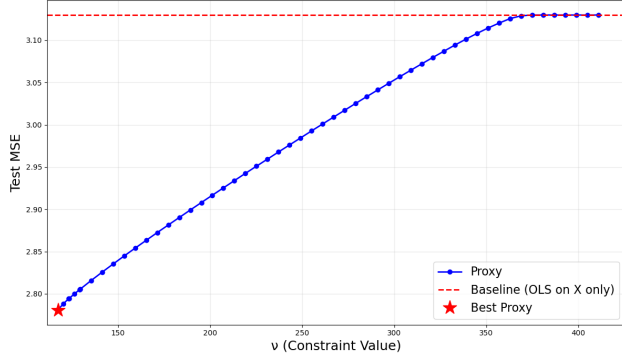


Figure 1. Performance when the privileged signal is strong (large $\|\theta\|_2$).

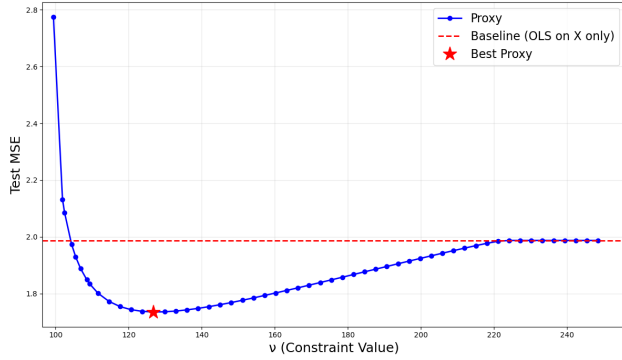


Figure 2. Performance when the privileged signal is weak (small $\|\theta\|_2$).

Henceforth, we identify any $f : \mathcal{X} \rightarrow \mathbb{R}$ with its lift $\tilde{f} : \mathcal{Z} \rightarrow \mathbb{R}$, defined by $\tilde{f}(x, w) = f(x)$, but we will later just write f for its lift as well.

Negative Transfer. The constraint level ν governs how much the rich-view model g is allowed to deviate from the labeled responses, and therefore how strongly privileged information can influence the deployment model f . In particular, feasibility requires

$$\nu \geq \min_{g \in \mathcal{G}} \frac{1}{n} \sum_{i=1}^n (Y_i - g(Z_i))^2.$$

If $\mathcal{F} \subseteq \mathcal{G}$ (identifying $f \in \mathcal{F}$ with its lift $\tilde{f}(x, w) = f(x)$), then choosing ν large enough that $g = f$ is feasible makes the unlabeled agreement term vanish, and the solution reduces to the labeled least-squares fit over $\mathcal{F}$. In contrast, choosing ν close to the minimum labeled risk over $\mathcal{G}$ forces g to be (approximately) the labeled least-squares fit over $\mathcal{G}$, yielding behavior akin to the aforementioned two-stage pseudo-labeling procedure. Thus, choosing ν too small can overemphasize noisy pseudo-responses and cause negative transfer, while choosing larger ν attenuates the influence of g and interpolates back toward supervised learning that ignores W .

To illustrate this behavior, we consider a linear model of the form

$$Y = \beta^\top X + \theta^\top W + \varepsilon,$$

where (X, W, ε) are jointly Gaussian, and the strength of the privileged feature is governed by the magnitude of $\|\theta\|_2$, while all other aspects of the data-generating process are held fixed.

Figures 1 and 2 illustrate the performance of the proposed method under varying levels of privileged signal strength. In Figure 1, where $\|\theta\|_2$ is large, the privileged variables are highly informative, and the optimal performance is achieved for small values of ν . In this regime, the method behaves like a two-stage procedure, leveraging the strong privileged signal to improve estimation accuracy.

In contrast, Figure 2 corresponds to a setting in which the privileged signal is weak. Here, aggressively incorporating pseudo-responses can degrade performance and even perform worse than the least-squares estimator (OLS) based only on the labeled pairs $\{(X_i, Y_i)\}_{i=1}^n$. This phenomenon of negative transfer was also noted in Xia & Wainwright (2024, Section 2.4.2) as a limitation of the two-stage approach.

Together, these results show that the proposed coupled training framework adaptively moderates the influence of the rich-view model, interpolating between regimes where privileged information is beneficial or detrimental, and thereby yields a robust, principled approach to leveraging privileged information.

1.2. Related Work

We place our coupled training framework in context by reviewing connections to learning using privileged information, semi-supervised learning (especially pseudo-labeling), multi-view agreement methods, and related optimization approaches.

Learning Using Privileged Information. Our setup is closely related to Learning Using Privileged Information (LUPI) (Vapnik & Vashist, 2009; Vapnik et al., 2015), where privileged variables W are available during training but the goal is to learn a predictor that depends only on X . However, in contrast to our framework, classical LUPI formulations assume fully labeled data and do not leverage unlabeled samples.

Semi-Supervised Learning. Learning from both labeled and unlabeled data has been widely studied across statistics and machine learning under the name semi-supervised learning (SSL) (Chapelle et al., 2006).

Among SSL methods, pseudo-labeling is a particularly prominent approach. In classification, pseudo-labeling was explored in the context of support vector machines

by Joachims (1999) and later extended to deep learning frameworks by Lee (2013). In regression, pseudo-responses were constructed via kernel smoothing and imputation in linear models (Chakraborty & Cai, 2018), extended to kernel ridge regression (Wang, 2023), and studied in general settings by Xia & Wainwright (2024). From an inferential perspective, mean estimation with SSL data was investigated by Zhang et al. (2019); Zhang & Bradic (2021), while Laine & Aila (2017) proposed an ensemble-based method that iteratively assigns pseudo-responses using historical predictions. A parallel line of work is weakly supervised learning, which relies on weak or noisy labels rather than ground truth. The Data Programming paradigm of Ratner et al. (2016; 2017) assumes access to labeling functions and was applied to MRI analysis by Fries et al. (2019). Weak observations were also used for data augmentation by Robinson et al. (2020), who established theoretical guarantees under universal central conditions. Closely related is the literature on learning with noisy labels (Natarajan et al., 2013; Song et al., 2022).

In contrast to these approaches, which primarily rely on pseudo-responses or weak supervision, our framework explicitly controls the influence of privileged information within an adaptive self-training scheme.

Multi-View Learning. The objective in (1) is related to the *co-regularized least-squares* objective of Sindhwani et al. (2005), which encourages predictors built from different views to agree on unlabeled data. In their multi-view SSL setting, one observes labeled examples $\{(X_i, Y_i)\}_{i=1}^n$ and unlabeled examples $\{X_i\}_{i=n+1}^N$, where each input $X_i = (X_i^{(1)}, X_i^{(2)})$ comes with two complementary representations $X_i^{(1)} \in \mathcal{X}^{(1)}$ and $X_i^{(2)} \in \mathcal{X}^{(2)}$. The goal is to learn $h = (h^{(1)}, h^{(2)})$ so that each component fits the labeled data in its own view, while their predictions are pushed to match on the unlabeled inputs (see Sindhwani et al. (2005, Section 3)). Related agreement-style objectives also appear in Ding et al. (2022), though their setting is fully supervised and does not use unlabeled data.

In our case, (1) uses a similar kind of agreement term, but for a different reason. We work in a *privileged-information* setting where W is available during training but not at deployment, so only $f(X)$ can be used at test time. That makes the coupling inherently asymmetric: f is the deployment model, and g is an auxiliary rich-view model used to pass information to f during training. We also take g to act on the joint input $Z = (X, W)$, rather than on W alone. This lets g capture effects of the privileged variables that depend on X (including X - W interactions) and helps translate whatever signal is present in W into improvements in f .

Greedy and Dictionary-Based Optimization. In Section 3, we restrict the deployment and rich-view models to

dictionary spans and fit them using an *alternating forward selection* procedure. This connects our method to the literature on greedy approximation in Hilbert spaces, where algorithms construct an additive predictor by sequentially selecting atoms that best reduce a squared-loss objective and enjoy deterministic approximation guarantees with sublinear objective or approximation error decay (Barron et al., 2008; DeVore & Temlyakov, 1996). It can also be viewed as a block-coordinate variant of classical forward (stepwise) selection in regression (Hastie et al., 2009, Sec. 3.3.2).

Our setting differs in that we perform greedy atom selection within each block while alternating between updating f with g fixed and updating g with f fixed, so the greedy search is carried out in a product space with interdependent residuals. Although support selection is combinatorial, we prove that the alternating greedy iterates achieve a global sublinear decay of the empirical coupled objective (Theorem 3.1), extending classical greedy-approximation analyses to the coupled privileged-information setting. This is complementary to alternating-minimization analyses for nonconvex problems such as dictionary learning, which typically focus on reconstruction or coefficient recovery under structural assumptions (Chatterji & Bartlett, 2017; Agarwal et al., 2014; Ruetz & Schnass, 2023); here the dictionaries are fixed and our guarantees target optimization error, and its translation to prediction error (risk), for the coupled estimator.

1.3. Notation

Let $\mathcal{F} \subset L^2(\mathcal{X})$ and $\mathcal{G} \subset L^2(\mathcal{Z})$ be closed, convex function classes. For a nonempty closed convex set C in a Hilbert space $(\mathcal{H}, \langle \cdot, \cdot \rangle)$, let $\Pi_C h \in \arg \min_{c \in C} \|h - c\|_{\mathcal{H}}^2$ denote the (metric) projection. When C is a closed linear subspace, Π_C coincides with the orthogonal projector.

For fixed points $x^N = (x_1, \dots, x_N) \in \mathcal{X}^N$ and $z^N = (z_1, \dots, z_N) \in \mathcal{Z}^N$, where $z_i = (x_i, w_i)$, let $\hat{P}_N^X$ and $\hat{P}_N^Z$ denote the corresponding normalized empirical measures. For functions $f : \mathcal{X} \rightarrow \mathbb{R}$ and $g : \mathcal{Z} \rightarrow \mathbb{R}$, we write $\|f\|_{L^p(\hat{P}_N^X)} := \left(\frac{1}{N} \sum_{i=1}^N |f(x_i)|^p \right)^{1/p}$, and $\|g\|_{L^p(\hat{P}_N^Z)} := \left(\frac{1}{N} \sum_{i=1}^N |g(z_i)|^p \right)^{1/p}$. For a class $\mathcal{F} \times \mathcal{G}$, define the max norm $\|(f, g)\|_{1, \infty} = \max\{\|f\|_{L^1(\hat{P}_N^X)}, \|g\|_{L^1(\hat{P}_N^Z)}\}$. An ϵ -cover of $\mathcal{F} \times \mathcal{G}$ with respect to $\|\cdot\|_{1, \infty}$ is a finite collection $\{(f_1, g_1), \dots, (f_M, g_M)\} \subset \mathcal{F} \times \mathcal{G}$ such that every $(f, g) \in \mathcal{F} \times \mathcal{G}$ is within ϵ of some (f_j, g_j) under this norm, and the minimal such M is denoted by $N_{1, \infty}(\epsilon, \mathcal{F} \times \mathcal{G}, z^N)$.

2. Risk Bounds

Lagrangian Relaxation. The constrained problem (1) admits a Lagrangian relaxation, leading to the penalized

objective

$$\hat{\mathcal{L}}(f, g; \lambda) := \frac{1}{N} \left(\sum_{i=1}^n (Y_i - f(X_i))^2 + \sum_{j=n+1}^N (g(Z_j) - f(X_j))^2 + \lambda \sum_{i=1}^n (Y_i - g(Z_i))^2 \right). \quad (2)$$

For an appropriate pairing (ν, λ) (e.g., when λ is dual-optimal and strong duality holds), the minimizers of (1) and (2) coincide. In particular, there exists $\tilde{\lambda} \geq 0$ and a minimizer $(\hat{f}, \hat{g})$ of (1) such that $(\hat{f}, \hat{g}) \in \arg\min_{(f, g)} \hat{\mathcal{L}}(f, g; \tilde{\lambda})$ and $\tilde{\lambda} \left(\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{g}(Z_i))^2 - \nu \right) = 0$. For simplicity, we focus throughout on the penalized formulation (2) and treat λ as a tuning parameter. Note that λ plays a role analogous to ν , but in the opposite direction. That is, smaller λ relaxes the effective constraint on g , attenuating the influence of pseudo-responses, while larger λ forces g to fit the labeled data more closely and increases the impact of pseudo-labeling. In particular, if $\mathcal{F} \subseteq \mathcal{G}$, as $\lambda \rightarrow 0$, the unlabeled data play no role and the solution reduces to the ordinary least-squares fit based only on the labeled data, whereas, as $\lambda \rightarrow \infty$, the solution approaches the two-stage procedure.

Population Minimizers. We now turn to a theoretical analysis of the proposed method and derive bounds on the prediction error of the learned deployment model $\hat{f}$. To this end, we consider the population-level counterpart of the empirical objective in (2), given by

$$\mathcal{L}(f, g; \lambda) := \frac{n}{N} \mathbb{E}[(Y - f(X))^2] + \frac{m}{N} \mathbb{E}[(g(Z) - f(X))^2] + \lambda \frac{n}{N} \mathbb{E}[(Y - g(Z))^2]. \quad (3)$$

Recall we identify any $f : X \rightarrow \mathbb{R}$ with its lift to $\mathcal{Z}$ via $\tilde{f}(x, w) = f(x)$. We define the rich-view regression function

$$\eta(z) := \mathbb{E}[Y \mid Z = z],$$

which is related to μ via $\mu(x) = \mathbb{E}[\eta(Z) \mid X = x]$.

Theorem 2.1 (Population Minimizers). *Any $(f^*, g^*) \in \mathcal{F} \times \mathcal{G}$ minimizing (3) satisfies the fixed-point equations:*

$$f^* = \Pi_{\mathcal{F}} \left(\frac{n}{N} \mu + \frac{m}{N} \mathbb{E}[g^*(Z) \mid X] \right), \\ g^* = \Pi_{\mathcal{G}} \left(\frac{m}{m + n\lambda} f^* + \frac{n\lambda}{m + n\lambda} \eta \right).$$

Furthermore, if $\mu \in \mathcal{F}$, $\mu \in \mathcal{G}$, and $\eta \in \mathcal{G}$, then

$$f^* = \mu, \text{ and } g^* = \frac{m}{m + n\lambda} \mu + \frac{n\lambda}{m + n\lambda} \eta.$$

Remark 2.2 (Interpolation Effect). The characterization in Theorem 2.1 highlights how the rich-view model g^* interpolates between the deployment target μ and the rich-view

regression function η . In particular, under the realizability assumptions in Theorem 2.1, as $\lambda \rightarrow 0$, the penalty on the rich-view model becomes inactive and the solution satisfies $g^* \rightarrow \mu$, corresponding to a regime in which the privileged information is ignored. On the other hand, as λ increases, g^* places increasing weight on η , recovering behavior akin to a two-stage pseudo-labeling procedure. Intermediate values of λ yield an interpolation between these extremes, allowing the method to adapt to the informativeness of the privileged variables.

Correlation-Controlled Risk Bound. We now study the statistical behavior of the proposed estimator by analyzing its prediction error. Let $(\hat{f}, \hat{g}) \in \mathcal{F} \times \mathcal{G}$ denote the solution (or an approximate solution) of the empirical problem (2) and let $(f^*, g^*) \in \mathcal{F} \times \mathcal{G}$ denote the minimizer of (3). Our goal is to control the excess risk

$$\mathbb{E}_{\mathcal{D}, X} [(\hat{f}(X) - \mu(X))^2],$$

where the expectation is over the training data $\mathcal{D}$ used to construct $\hat{f}$ and an independent test point X .

A key quantity in our analysis is a correlation coefficient that captures the interaction between the estimation errors of the deployment and rich-view models. Specifically, define the estimation errors

$$\hat{e}_f(X) := f^*(X) - \hat{f}(X), \quad \hat{e}_g(Z) := g^*(Z) - \hat{g}(Z).$$

We then define

$$\rho_{\star} = \frac{|\mathbb{E}_{\mathcal{D}, Z} [\hat{e}_f(X) \hat{e}_g(Z)]|}{(\mathbb{E}_{\mathcal{D}, X} [\hat{e}_f^2(X)])^{1/2} (\mathbb{E}_{\mathcal{D}, Z} [\hat{e}_g^2(Z)])^{1/2}} \in [0, 1],$$

where $Z = (X, W)$, which measures the alignment between the residuals of the two estimators.

The following corollary, which is a direct consequence of the excess loss decomposition in Theorem C.1, shows that $\rho_{\star}$ plays a central role in controlling the prediction error.

Corollary 2.3 (Correlation-Controlled Risk Bound). *If $\mu \in \mathcal{F}$, $\mu \in \mathcal{G}$ (note that functions on X are implicitly lifted to $\mathcal{Z}$), and $\eta \in \mathcal{G}$, then any estimator $(\hat{f}, \hat{g}) \in \mathcal{F} \times \mathcal{G}$ satisfies*

$$\mathbb{E}_{\mathcal{D}, X} [(\hat{f}(X) - \mu(X))^2] \leq \frac{\mathbb{E}_{\mathcal{D}} [\mathcal{L}(\hat{f}, \hat{g}; \lambda) - \mathcal{L}(f^*, g^*; \lambda)]}{\gamma_{n, m, \lambda}(\rho_{\star})},$$

where

$$\gamma_{n, m, \lambda}(\rho_{\star}) := 1 - \frac{m^2}{N(m + n\lambda)} \rho_{\star}^2 \geq \frac{n}{N}.$$

The above bound shows that excess loss in the joint objective translates into prediction error for $\hat{f}$, with a constant that degrades as the correlation $\rho_{\star}$ between the estimation errors increases.

Remark 2.4 (Correlation Score Interpretation). (i) The correlation score $\rho_\star \in [0, 1]$ quantifies how strongly the residuals of the rich-view estimator $\hat{g}$ and the deployment estimator $\hat{f}$ are dependent. When $\rho_\star$ is small, the two models make largely independent errors, indicating that the privileged features W contain information about Y that is not captured by X . In this regime, leveraging the privileged information can substantially improve the accuracy of $\hat{f}$. In contrast, when $\rho_\star$ is large, the residuals of $\hat{f}$ and $\hat{g}$ are highly correlated, suggesting that W provides little additional information beyond what is already contained in X , and the benefit of incorporating privileged data becomes limited. (ii) By Cauchy–Schwarz and the law of total variance, it can be shown (see Section C.4 for details) that

$$\rho_\star^2 \leq \frac{\mathbb{E}_{\mathcal{D}, X} \left[(\mathbb{E}[\hat{e}_g(Z) | \mathcal{D}, X])^2 \right]}{\mathbb{E}_{\mathcal{D}, Z} [\hat{e}_g^2(Z)]} \leq 1.$$

Furthermore, if $\mu \in \mathcal{F}$, $\mu \in \mathcal{G}$, and $\eta \in \mathcal{G}$, then $\mathbb{E}[g^\star(Z) | X] = \mu(X)$, and hence

$$\mathbb{E}[\hat{e}_g(Z) | \mathcal{D}, X] = \mu(X) - \mathbb{E}[\hat{g}(Z) | \mathcal{D}, X].$$

Consequently, $\rho_\star$ is small when the conditional rich-view model error $\mathbb{E}_{\mathcal{D}, X} [\mathbb{E}[\hat{e}_g(Z) | \mathcal{D}, X]^2]$ is small relative to the total rich-view model error $\mathbb{E}_{\mathcal{D}, Z} [\hat{e}_g^2(Z)]$. This corresponds to a regime in which the rich-view model $\hat{g}$ leverages privileged information W to capture predictive signal about Y that is not well explained by X alone, thereby reducing the component of the rich-view error that is aligned with X . Hence, $\rho_\star$, and consequently the risk bound, are controlled by a *multiplicative* relative error, in contrast to [Xia & Wainwright \(2024\)](#), where the risk is bounded in terms of an *additive* absolute error $\mathbb{E}_{\mathcal{D}, X} [\mathbb{E}[\hat{e}_g(Z) | \mathcal{D}, X]^2]$. This means that the risk bound more gracefully degrades as $\hat{g}$ worsens, because it depends on the fraction of rich-view error explainable from X rather than on its absolute size.

2.1. Main Risk Bound

Assumption 2.5. The function classes $\mathcal{F}$ and $\mathcal{G}$ are uniformly bounded. Specifically, there exists a constant $B > 0$ such that

$$\sup_{(f, g) \in \mathcal{F} \times \mathcal{G}} \|(f, g)\|_\infty \leq B \quad \text{and} \quad |Y| \leq B, \quad \text{a.s.,}$$

where $\|(f, g)\|_\infty := \max \{\|f\|_\infty, \|g\|_\infty\}$.

We introduce a complexity term that captures the joint richness of the deployment and rich-view function classes, at scales determined by the sample sizes and the regularization parameter λ . Define

$$\mathfrak{E}_{\mathcal{F} \times \mathcal{G}}^{N, \lambda} := \sup_{z^N} \log \left(\mathcal{N}_{1, \infty}(\epsilon_N, \mathcal{F} \times \mathcal{G}, z^N) \right),$$

$$\text{with } \epsilon_N = \frac{1}{160B(\lambda+1)^{3/2}N}.$$

Theorem 2.6 (Risk Bound). If $\mu \in \mathcal{F}$, $\mu \in \mathcal{G}$, $\eta \in \mathcal{G}$ and [Assumption 2.5](#) holds, then any estimator $(\hat{f}, \hat{g}) \in \mathcal{F} \times \mathcal{G}$ satisfies

$$\begin{aligned} & \mathbb{E}_{\mathcal{D}, X} [(\hat{f}(X) - \mu(X))^2] \\ & \leq \frac{1}{\gamma_{n, m, \lambda}(\rho_\star)} \left(\frac{C_1(\lambda + 1) \mathfrak{E}_{\mathcal{F} \times \mathcal{G}}^{N, \lambda}}{N} + 2\mathbb{E}_{\mathcal{D}} [\hat{\Delta}(\hat{f}, \hat{g})] \right), \end{aligned} \quad (4)$$

where $C_1 > 0$ is a universal constant, and

$$\hat{\Delta}(\hat{f}, \hat{g}) := \hat{\mathcal{L}}(\hat{f}, \hat{g}; \lambda) - \hat{\mathcal{L}}(f^\star, g^\star; \lambda).$$

Remark 2.7 (Optimization Error). (i) The bound separates the estimation error into a complexity term controlled by the joint richness of $\mathcal{F}$ and $\mathcal{G}$, and an *optimization error* term measuring excess empirical risk relative to $(f^\star, g^\star)$. (ii) If $(\hat{f}, \hat{g})$ are exact minimizers of (2), then

$$\hat{\Delta}(\hat{f}, \hat{g}) = \hat{\mathcal{L}}(\hat{f}, \hat{g}; \lambda) - \hat{\mathcal{L}}(f^\star, g^\star; \lambda) \leq 0,$$

and the optimization error term in (4) can be dropped. However, when $(\hat{f}, \hat{g})$ are obtained from an approximate optimization procedure, for instance, from the alternating coupled training algorithm, the excess empirical loss $\hat{\Delta}(\hat{f}, \hat{g})$ must be explicitly controlled. As an example, in [Theorem 3.1](#) below, we show that when $\mathcal{F}$ and $\mathcal{G}$ are linear spans of dictionaries $\mathcal{D}_f$ and $\mathcal{D}_g$, the iterates $(\hat{f}, \hat{g}) = (f_k, g_k)$ of an alternating forward selection algorithm satisfy

$$\begin{aligned} & \hat{\Delta}(\hat{f}, \hat{g}) \\ & \leq C_2 \frac{\max\{1, \lambda\} (\|f^\star\|_{L^1(\mathcal{D}_f)} + \|g^\star\|_{L^1(\mathcal{D}_g)})^2 \log(k+1)}{k}, \end{aligned}$$

for some universal constant $C_2 > 0$.

Remark 2.8 (Effect of Correlation Score). (i) Note that, for any values of $\rho_\star$ and λ , the quantity $\gamma_{n, m, \lambda}(\rho_\star)$ satisfies the uniform lower bound $\gamma_{n, m, \lambda}(\rho_\star) \geq n/N$. As a consequence,

$$\frac{1}{\gamma_{n, m, \lambda}(\rho_\star)} \cdot \frac{1}{N} \leq \frac{1}{n},$$

so the $O(1/N)$ term in (4) is never worse than the $O(1/n)$ dependency obtained from using only the labeled data.

(ii) When the estimation errors of the deployment and rich-view estimators are highly correlated (i.e., $\rho_\star$ is close to 1), one has $\gamma_{n, m, \lambda}(\rho_\star) \approx 1 - m^2/(N(m + n\lambda))$. In particular, when $\lambda = 0$, this becomes $\gamma_{n, m, 0}(\rho_\star) \approx n/N$, and hence

$$\frac{1}{\gamma_{n, m, 0}(\rho_\star)} \cdot \frac{1}{N} \approx \frac{1}{n},$$

so the $O(1/N)$ term in (4) recovers the $O(1/n)$ dependence, obtained when learning solely from n labeled observations.

(iii) More generally, $\gamma_{n, m, \lambda}(\rho_\star)$ controls the multiplicative constant in (4) through its dependence on $\rho_\star$. When the deployment and rich-view estimation errors are weakly aligned (small $\rho_\star$), $\gamma_{n, m, \lambda}(\rho_\star)$ remains close to one, so the $O(1/N)$ term is not significantly inflated.

3. High-Dimensional Extension

In Section 1.1, we proposed an alternating coupled training procedure for jointly estimating the deployment model $\hat{f}$ and the rich-view model $\hat{g}$. In practice, however, directly optimizing over arbitrary functions in infinite-dimensional Hilbert spaces is computationally inefficient, especially when the combined inputs (X, W) are high-dimensional.

To bridge this gap between theory and practice, we adopt a dictionary-based approximation strategy in which functions are represented as sparse linear combinations of elementary functions, or *atoms*, drawn from predefined dictionaries. The resulting algorithms are efficient and remain theoretically well-justified. In this section, we introduce a greedy alternating forward selection procedure that incrementally constructs approximations of $(\hat{f}, \hat{g})$ using finite dictionaries.

Let $\mathcal{D}_f$ and $\mathcal{D}_g$ be finite, symmetric dictionaries such that $\|\psi\|_{L^2(\hat{P}_N^X)} \leq 1$ for all $\psi \in \mathcal{D}_f$ and $\|\phi\|_{L^2(\hat{P}_N^Z)} \leq 1$ for all $\phi \in \mathcal{D}_g$. For functions $f \in \mathcal{F} = \text{span}(\mathcal{D}_f)$ and $g \in \mathcal{G} = \text{span}(\mathcal{D}_g)$ of the form

$$f = \sum_{\psi \in \mathcal{D}_f} c_\psi \psi, \quad g = \sum_{\phi \in \mathcal{D}_g} c_\phi \phi,$$

we define the associated atomic norms by

$$\|f\|_{L^1(\mathcal{D}_f)} := \inf \left\{ \sum_{\psi \in \mathcal{D}_f} |c_\psi| : f = \sum_{\psi \in \mathcal{D}_f} c_\psi \psi \right\},$$

$$\|g\|_{L^1(\mathcal{D}_g)} := \inf \left\{ \sum_{\phi \in \mathcal{D}_g} |c_\phi| : g = \sum_{\phi \in \mathcal{D}_g} c_\phi \phi \right\}.$$

3.1. Alternating Forward Selection Algorithm

We now describe a greedy algorithm that alternates between updating f and recalibrating g .

It approximately minimizes $\hat{\mathcal{L}}(f, g; \lambda)$ in (2) by alternately updating f and g using forward (stepwise) selection (Hastie et al., 2009, Sec. 3.3.2).

Alternating Forward Selection Algorithm. Initialize $g_0 = 0$. For $k = 1, 2, \dots$, alternate between:

1. **Update f .** Given the current rich-view model $\hat{Y} = g_{k-1}$, update f by solving

$$(\psi_k, f_k) \in \underset{\substack{\psi \in \mathcal{D}_f \\ f \in \text{span}(\psi_1, \dots, \psi_{k-1}, \psi)}}{\text{argmin}} \frac{1}{N} \left(\sum_{i=1}^n (Y_i - f(X_i))^2 + \sum_{j=n+1}^N (g_{k-1}(Z_j) - f(X_j))^2 \right).$$

2. **Recalibrate g .** Given the current deployment model f_k , update g by solving

$$(\phi_k, g_k) \in \underset{\substack{\phi \in \mathcal{D}_g \\ g \in \text{span}(\phi_1, \dots, \phi_{k-1}, \phi)}}{\text{argmin}} \frac{1}{N} \left(\sum_{j=n+1}^N (g(Z_j) - f_k(X_j))^2 + \lambda \sum_{i=1}^n (Y_i - g(Z_i))^2 \right).$$

An efficient implementation of this procedure is provided in Appendix A (see Algorithm 1). Before turning to implementation details, we establish a convergence rate for the estimators obtained after k iterations.

Theorem 3.1 (Alternating Forward Selection Convergence Rate). *For all $k \geq 1$, and all $f \in \text{span}(\mathcal{D}_f)$ and $g \in \text{span}(\mathcal{D}_g)$, the iterates $(\hat{f}, \hat{g}) = (f_k, g_k)$ produced by the Alternating Forward Selection Algorithm satisfy*

$$\hat{\mathcal{L}}(\hat{f}, \hat{g}; \lambda) \leq \hat{\mathcal{L}}(f, g; \lambda) + C_2 \frac{\max\{1, \lambda\} (\|f\|_{L^1(\mathcal{D}_f)} + \|g\|_{L^1(\mathcal{D}_g)})^2 \log(k+1)}{k},$$

for a universal constant $C_2 > 0$.

This result shows that the greedy alternating procedure achieves a sublinear optimization error. Moreover, when the function classes $\mathcal{F}$ and $\mathcal{G}$ are restricted to the spans of dictionaries $\mathcal{D}_f$ and $\mathcal{D}_g$, respectively, the statistical complexity term in Theorem 2.6 admits an explicit bound in terms of the number of greedy iterations. Specifically, if $(\hat{f}, \hat{g}) = (f_k, g_k)$ lie in $\text{span}(\mathcal{D}_f) \times \text{span}(\mathcal{D}_g)$ with k selected atoms in each component, then applying Barron et al. (2008, Lemma 3.3) to the associated covering numbers yields

$$\mathfrak{N}_{\mathcal{F} \times \mathcal{G}}^{N, \lambda} \leq (k+1) \log(C_3 B(\lambda+1)^{3/2} N),$$

for some universal constant $C_3 > 0$.

Consequently, under the assumptions of Theorem 2.6, the iterates $(\hat{f}, \hat{g}) = (f_k, g_k)$ produced by Algorithm 1 satisfy

$$\mathbb{E}_{\mathcal{D}, X} [(\hat{f}(X) - \mu(X))^2] \leq \frac{1}{\gamma_{n,m,\lambda}(\rho_\star)} \left(\frac{C_1(\lambda+1)(k+1) \log(C_3 B(\lambda+1)^{3/2} N)}{N} + \frac{2C_2 \max\{1, \lambda\} (\|f^\star\|_{L^1(\mathcal{D}_f)} + \|g^\star\|_{L^1(\mathcal{D}_g)})^2 \log(k+1)}{k} \right).$$

3.2. Efficient Implementation

We briefly describe an efficient implementation of the Alternating Forward Selection Algorithm; full derivations and pseudocode are deferred to Appendix A. At a high level,

each block update (the f -step over $\text{span}(\mathcal{D}_f)$ and the g -step over $\text{span}(\mathcal{D}_g)$) is a least-squares refit over the currently selected atoms. Rather than resolving these refits from scratch, we maintain incremental QR decompositions for the (weighted) design matrices associated with the selected atoms, enabling $O(Nk)$ updates per iteration, where k is the number of selected atoms per block.

Time complexity. Assuming each candidate atom can be evaluated on all N samples in $O(N)$ time, selecting the next atom at iteration k amounts to scanning the remaining candidates in the relevant dictionary and computing their correlations with the current residual after orthogonalization against the selected span (via the maintained QR factors). This yields a total cost up to step k of

$$O(N(|\mathcal{D}_f| + |\mathcal{D}_g|)k).$$

For comparison, a naive version of forward selection over paired atoms (ψ, ϕ) (i.e., scanning a product dictionary) scales as

$$O(N|\mathcal{D}_f||\mathcal{D}_g|k),$$

which is substantially larger. As shown in Theorem 3.1, this computational improvement comes with only a $\log(k)$ factor difference in the optimization guarantee relative to classical greedy schemes Barron et al. (2008); DeVore & Temlyakov (1996).

4. Experiments

We evaluate our coupled training method on synthetic data and UCI medical regression benchmarks. Across all experiments, we observe that test MSE follows a U-shaped curve as a function of the Lagrangian multiplier λ in (2), so we treat λ as a hyperparameter and select it via cross-validation. We compare against two baselines: (i) ordinary least-squares using only labeled pairs $\{(X_i, Y_i)\}_{i=1}^n$ (Baseline), and (ii) the two-stage pseudo-labeling procedure from Section 1 (Two-Stage).

Our first experiment uses the UCI Parkinson’s Telemonitoring dataset to predict disease severity from voice measurements. We split the acoustic features so that basic perturbation measures (jitter, shimmer) serve as deployment features X , and complex spectral metrics that require careful calibration are treated as privileged features W . As shown in Figure 3, test MSE is minimized at an intermediate value of λ , and our method outperforms both Baseline and Two-Stage.

Our second experiment uses the UCI Cleveland Heart Disease dataset. Here, deployment features X include routine clinical measurements (age, blood pressure, cholesterol), while privileged features W include specialized diagnostic tests like fluoroscopy and thallium stress tests that are too costly for routine screening. Figure 4 shows that both Linear

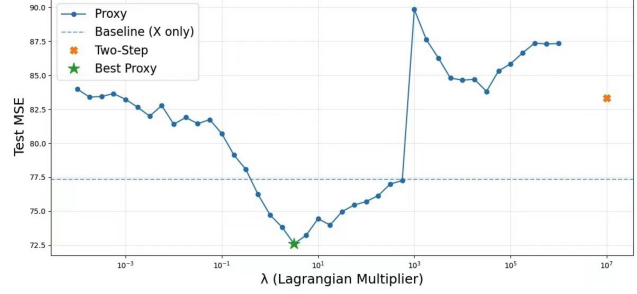


Figure 3. Parkinson’s dataset: U-shaped test MSE vs. λ . Setup: $n = 865$, $m = 4298$, $n_{\text{test}} = 712$; $d_X = 10$, $d_W = 9$; model: Random Forest.

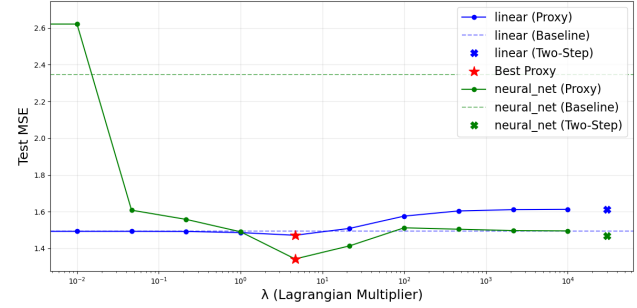


Figure 4. Heart dataset: U-shaped test MSE vs. λ . Setup: $n = 34$, $m = 166$, $n_{\text{test}} = 80$; $d_X = 10$, $d_W = 3$; models: Linear and Neural Net.

and Neural Net models exhibit U-shaped test MSE curves, with intermediate λ yielding the lowest error.

These results confirm the theoretical prediction that λ interpolates between ignoring privileged information ($\lambda \rightarrow 0$) and the Two-Stage procedure ($\lambda \rightarrow \infty$). In both cases, cross-validation selects an intermediate value that improves over the baselines (Table 1). We provide further experiments on synthetic linear data and on PneumoniaMNIST using Algorithm 1 in Appendix B.

Table 1. Cross-validated test MSE. $\hat{\lambda}$ selected via cross-validation on the labeled training set.

Dataset	Model	Baseline	Two-Stage	Ours	$\hat{\lambda}$
Parkinson’s	Random Forest	77.34	83.32	72.06	5.62
Heart	Linear	1.49	1.61	1.47	4.64
Heart	Neural Net	2.35	1.47	1.41	21.54

Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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A. Efficient Implementation of Alternating Forward Selection

Recall from Section 1 that we observe n labeled and m unlabeled samples with $N = n + m$, and that $\mathcal{Z} = \mathcal{X} \times \mathcal{W}$. Let $\pi_X : \mathcal{Z} \rightarrow \mathcal{X}$ be the canonical projection $\pi_X(x, w) = x$. Define the finite measures on $\mathcal{Z}$ by

$$\tilde{P}_n^{(1)} := \sum_{i=1}^n \delta_{Z_i}, \quad \tilde{P}_m^{(2)} := \sum_{j=n+1}^N \delta_{Z_j}, \quad \tilde{P}_n^{(3)} := \lambda \tilde{P}_n^{(1)},$$

and normalize by N :

$$\hat{P}_N^{(\ell)} := \frac{1}{N} \tilde{P}^{(\ell)}, \quad \ell \in \{1, 2, 3\}.$$

Note that $\hat{P}_N^{(1)} + \hat{P}_N^{(2)} = \hat{P}_N^Z$. We will work exclusively with Z -indexed norms by viewing the deployment class as a subspace of functions on $\mathcal{Z}$. Define the lifted deployment space

$$\mathcal{H}_f := \{f \in L^2(\hat{P}_N^Z) : f \text{ is } \sigma(\pi_X)\text{-measurable}\},$$

i.e., $f \in \mathcal{H}_f$ implies $f(x, w)$ depends only on x (equivalently, there exists $\tilde{f} : \mathcal{X} \rightarrow \mathbb{R}$ such that $f(x, w) = \tilde{f}(x)$ $\hat{P}_N^Z$ -a.e.).

Let $\mathcal{Q}_g := \hat{P}_N^{(2)} + \hat{P}_N^{(3)}$ and set

$$\mathcal{H}_g := L^2(\mathcal{Q}_g).$$

With this convention, for $f \in \mathcal{H}_f$, $g \in \mathcal{H}_g$, the empirical objective can be written as

$$\hat{\mathcal{L}}(f, g; \lambda) := \|Y - f\|_{L^2(\hat{P}_N^{(1)})}^2 + \|f - g\|_{L^2(\hat{P}_N^{(2)})}^2 + \|Y - g\|_{L^2(\hat{P}_N^{(3)})}^2.$$

We simplify the problem by considering the product space $\mathcal{H}^* := \mathcal{H}_f \times \mathcal{H}_g$ with inner product

$$\langle (u, v), (u', v') \rangle_{\star} := \langle u, u' \rangle_{L^2(\hat{P}_N^{(1)})} + \langle u - v, u' - v' \rangle_{L^2(\hat{P}_N^{(2)})} + \langle v, v' \rangle_{L^2(\hat{P}_N^{(3)})}. \quad (5)$$

This implies that, given a target (Y, Y) , minimizing $\hat{\mathcal{L}}(f, g; \lambda)$ is equivalent to minimizing

$$\hat{\mathcal{L}}(f, g; \lambda) = \|(Y, Y) - (f, g)\|_{\star}^2 \quad (6)$$

over $(f, g) \in \mathcal{H}^*$. Note that although we write the target as (Y, Y) , the unlabeled term involves $(Y - f) - (Y - g) = g - f$, so it does not depend on Y and therefore requires no labels for unlabeled samples. Without loss of generality, we suppose that the dictionaries $\mathcal{D}_f$ and $\mathcal{D}_g$ are bounded by 1 in the *pooled* empirical norm, i.e.,

$$\sup_{\psi \in \mathcal{D}_f} \|\psi\|_{L^2(\hat{P}_N^Z)} \leq 1, \quad \sup_{\phi \in \mathcal{D}_g} \|\phi\|_{L^2(\hat{P}_N^Z)} \leq 1.$$

In the following, we use canonical embeddings to construct the dictionary for the product Hilbert space $\mathcal{H}^*$ based on $\mathcal{D}_f$ and $\mathcal{D}_g$. Define embedded versions:

$$\mathcal{D}_f^* := \{(\psi, 0) : \psi \in \mathcal{D}_f\}, \quad \mathcal{D}_g^* := \{(0, \phi) : \phi \in \mathcal{D}_g\}.$$

Assume the union $\mathcal{D}^* := \mathcal{D}_f^* \cup \mathcal{D}_g^*$ is balanced, i.e., whenever $a \in \mathcal{D}^*$, we have $-a \in \mathcal{D}^*$. Since $\hat{P}_N^{(1)} + \hat{P}_N^{(2)} = \hat{P}_N^Z$, we have

$$\|(\psi, 0)\|_{\star}^2 = \|\psi\|_{L^2(\hat{P}_N^{(1)})}^2 + \|\psi\|_{L^2(\hat{P}_N^{(2)})}^2 = \|\psi\|_{L^2(\hat{P}_N^Z)}^2 \leq 1.$$

Moreover,

$$\|(0, \phi)\|_{\star}^2 = \|\phi\|_{L^2(\hat{P}_N^{(2)})}^2 + \|\phi\|_{L^2(\hat{P}_N^{(3)})}^2 = \frac{1}{N} \sum_{j=n+1}^N \phi(Z_j)^2 + \frac{\lambda}{N} \sum_{i=1}^n \phi(Z_i)^2 \leq \max\{1, \lambda\} \|\phi\|_{L^2(\hat{P}_N^Z)}^2 \leq \max\{1, \lambda\}.$$

Denote $R_\lambda := \max\{1, \sqrt{\lambda}\}$. Thus every embedded atom satisfies $\|a\|_{\star} \leq R_\lambda$. Throughout this subsection, Π_S and $\Pi_S^\perp$ denote the orthogonal projections onto a subspace $S \subset \mathcal{H}^*$ and its orthogonal complement, respectively, with respect to the inner product $\langle \cdot, \cdot \rangle_{\star}$.

Algorithm 1 Alternating Forward Selection

Input: Dictionaries $\mathcal{D}_f^*, \mathcal{D}_g^*$; initial values $(f_0, g_0) = (0, 0)$, $S_0^f = S_0^g = \{0\}$; target (Y, Y) .

for $k = 1, 2, \dots$ **do**

 Compute the residual:

$$r_k \leftarrow (Y, Y) - (f_{k-1}, g_{k-1}).$$

Step 1: f-selection and projection

 Select

$$\psi_k^* = (\psi_k, 0) \in \mathcal{D}_f^*$$

 maximizing

$$a \mapsto \frac{\langle r_k, \Pi_{S_{k-1}^f}^\perp a \rangle_\star}{\|\Pi_{S_{k-1}^f}^\perp a\|_\star}.$$

 Update

$$S_k^f \leftarrow \text{span}(\psi_1^*, \dots, \psi_k^*),$$

$$(f_k, g_{k-1}) \leftarrow (f_{k-1}, g_{k-1}) + \Pi_{S_k^f} r_k,$$

 and set

$$r_k^{(g)} \leftarrow r_k - \Pi_{S_k^f} r_k.$$

Step 2: g-selection and projection

 Select

$$\phi_k^* = (0, \phi_k) \in \mathcal{D}_g^*$$

 maximizing

$$b \mapsto \frac{\langle r_k^{(g)}, \Pi_{S_{k-1}^g}^\perp b \rangle_\star}{\|\Pi_{S_{k-1}^g}^\perp b\|_\star}.$$

 Update

$$S_k^g \leftarrow \text{span}(\phi_1^*, \dots, \phi_k^*),$$

$$(f_k, g_k) \leftarrow (f_k, g_{k-1}) + \Pi_{S_k^g} r_k^{(g)},$$

end for

B. Implementation Details and Additional Experiments

B.1. Parkinson’s (Telemonitoring)

This experiment targets two practical challenges in medical deployment: (i) limited labeled outcomes (ground-truth $Y = \text{total_UPDRS}$ requires clinical assessment), and (ii) degraded signal fidelity in real-world recordings. We use the Parkinson’s Telemonitoring dataset from the UCI repository, where each row corresponds to one voice measurement and each subject (subject#) may have multiple recordings.

Feature sets. We partition covariates into deployment features X and privileged features W to model the fact that some high-fidelity acoustic descriptors are only reliable in controlled clinical conditions:

- **Deployment features** X (10): {age, sex, test_time, Jitter(%), Jitter(Abs), Jitter:RAP, Jitter:PPQ5, Jitter:DDP, Shimmer, Shimmer(dB)}.
- **Privileged features** W (9): {Shimmer:APQ3, Shimmer:APQ5, Shimmer:APQ11, Shimmer:DDA, NHR, HNR, RPDE, DFA, PPE}.

Intuitively, W contains detailed perturbation/spectral measures that can be informative when recorded under standardized

settings, but can be unreliable on consumer devices due to background noise and gain variation; X is comparatively more robust. Our goal is to exploit W during training to improve a predictor $\hat{f}(X)$ that only uses X at deployment.

Group-wise splitting (no subject leakage). To prevent leakage across repeated measurements from the same individual, we perform a single `GroupShuffleSplit` by `subject#` into train/test with `test_size=0.1` and fixed seed (reported runs use `seed = 202`). Inside the training subjects, we further split `subjects` into labeled vs. unlabeled sets: a fraction `frac=0.15` of training subjects are labeled and the rest are treated as unlabeled. This ensures that labeled/unlabeled/test are disjoint by `subject`.

Preprocessing and model. All preprocessing is fit on training data only, the test set is never used for fitting. We implement a `ColumnTransformer`: (i) “voice” features (all non-demographic, non-sex covariates) are standardized and then reduced by PCA with $n = 12$ components (clipped to the available dimension if fewer than 12 features), (ii) demographics `{age, test_time}` are standardized, (iii) sex is passed through unchanged. As the base regressor, we use `RandomForestRegressor` with fixed hyperparameters: `n_estimators=1400`, `max_depth=12`, `min_samples_leaf=4`, `max_features=sqrt`, `bootstrap=True`, and the same random seed.

Target scaling and evaluation. We standardize the labeled outcomes Y using a `StandardScaler` fit *only* on labeled targets. All models are trained to predict scaled Y ; at evaluation we inverse-transform predictions back to the original UPDRS scale, clip to $[0, 100]$, and report test mean squared error (MSE).

Baselines and proxy learning. **Baseline** trains f on labeled data only using $X \rightarrow Y$. **Two-step** trains a teacher on labeled data using $(X, W) \rightarrow Y$, pseudo-labels unlabeled data, then trains a student on $(X_L \cup X_U) \rightarrow (Y_L \cup \tilde{Y}_U)$. **Proxy learning** alternates updates of f and g for up to 15 iterations. For each λ in a log-spaced grid $\{10^{-4}, \dots, 10^6\}$ with 41 points, we initialize f from the labeled baseline and initialize g by fitting to a constant target $g_0 \equiv 0$. Each g -update uses sample weights λ on labeled points and weight 1 on unlabeled points. We employ early stopping based on stabilization of the disagreement on unlabeled data (patience=2, relative tolerance 5×10^{-5} , and a small prediction-change threshold). We then evaluate the resulting $\hat{f}(X)$ on the held-out test set.

As shown in Figure 3, the test MSE exhibits a U-shaped dependence on λ , attaining its minimum at an intermediate value and improving over both the baseline and two-step methods.

Cross Validation. For selecting $\hat{\lambda}$, we perform 5-fold `GroupKFold` cross-validation on the labeled set only (grouped by `subject#`); in each fold, the unlabeled pool U is kept fixed and used during training, while validation is performed on the labeled fold. We sweep $\lambda \in \{\logspace(-4, 6)\}$ with 41 points and report $\hat{\lambda}$ on the original scale. Proxy learning alternates updates of f and g with labeled weights λ and unlabeled weights 1, using early stopping (patience=2, relative tolerance 5×10^{-5} , prediction-change threshold 10^{-5}) and a λ -dependent iteration cap (`effective_max_iter`).

B.2. Heart Disease

This experiment targets a common deployment constraint in clinical screening: highly predictive tests are available during development but are impractical at scale due to cost, invasiveness, or equipment requirements. We use the UCI Heart Disease (Cleveland) dataset (`processed.cleveland.data`), where each row corresponds to one patient and the regression target is disease severity `num` $\in \{0, 1, 2, 3, 4\}$.

Feature sets. We partition covariates into deployment features X and privileged features W to reflect realistic test availability:

- **Deployment features X (10):** `{age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, slope}`.
- **Privileged features W (3):** `{oldpeak, ca, thal}`.

Here `ca` (fluoroscopy) and `thal` (stress/nuclear imaging) are costly/invasive, while X contains routine history/vitals. Our goal is to leverage W during training to improve a deployment predictor $\hat{f}(X)$ that only uses X .

Splitting (labeled/unlabeled/test). We drop rows with missing values (entries marked as “?” are treated as NaN and removed), then perform a single random shuffle with fixed seed (`seed=42`). We take `n_labeled=34` as labeled training points, then `n_unlabeled=166` as unlabeled training points (clipped to ensure enough remaining samples), and finally `n_test=80` as the held-out test set. All reported metrics are computed on this disjoint test set.

Preprocessing and models. To avoid test leakage, we standardize X and W using `StandardScaler` fit only on training inputs (labeled+unlabeled), then transform labeled/unlabeled/test accordingly. We evaluate two function classes: (i) **Linear** (`LinearRegression`); and (ii) **Neural net** (`MLPRegressor`) with hidden layers (32, 16), `max_iter=2000`, `early_stopping=True`, `validation_fraction=0.1`, and the same random seed.

Evaluation. We report test mean squared error (MSE) on the raw target scale (`num` in $\{0, \dots, 4\}$). No target scaling or clipping is applied.

We sweep $\lambda \in \{0\} \cup \text{logspace}(-2, 4, 10)$. For the linear case, we solve the corresponding convex problem with `cvxpy` using `OSQP` (`max_iter=10000`). For the neural-net case, we alternate f - and g -updates for up to 15 outer iterations: f fits on labeled (X_L, Y_L) plus unlabeled ($X_U, g(X_U, W_U)$), while g fits on labeled plus unlabeled ($X_U, f(X_U)$) with sample weights λ on labeled and 1 on unlabeled. We initialize g by fitting to a constant target (the labeled mean), and stop early when the objective stabilizes (absolute or relative change below 10^{-4}).

B.3. Synthetic Linear Data

Dimensions and sample sizes. Deployment features X have dimension $d_X = 5$ and privileged features W have dimension $d_W = 30$. Each run uses $n = 150$ labeled samples, $m = 6000$ unlabeled samples, and $n_{\text{test}} = 2000$ test samples. Noise standard deviation is $\sigma = 1$.

Data generation. For each dataset, we sample $X \in \mathbb{R}^{n \times d_X}$ i.i.d. from a standard normal distribution and include an explicit intercept term in all linear predictors. To control the dependence between X and the privileged features $W \in \mathbb{R}^{n \times d_W}$, we draw a random mixing matrix $A \in \mathbb{R}^{d_X \times d_W}$ and rescale each column to have ℓ_2 -norm $\sqrt{d_X}$. We then generate W from X using a correlation-strength parameter $\rho \in [0, 1]$ (equivalently controlling the signal–noise tradeoff in W):

$$\text{correlation.type} = \text{'linear'} : \quad W = \rho(XA) + \sqrt{1 - \rho^2}Z, \quad Z \sim \mathcal{N}(0, I_{d_W}) \text{ i.i.d.}$$

A single ground-truth parameter vector $\theta_\star \in \mathbb{R}^{1+d_X+d_W}$ is sampled once per run (intercept + X -block + W -block), and the coefficients corresponding to W are downscaled by a factor 0.2 to make the privileged block weaker. The labeled, unlabeled, and test sets share the same $\theta_\star$ and differ only by resampling (X, W) and observation noise, using random seeds `seed`, `seed+1`, and `seed+2`, respectively.

Proxy simulation. We implement the proxy method as a constrained convex program in `CVXPY` with two linear predictors: (i) an X -only deployment predictor β (with explicit intercept), and (ii) a rich-view predictor γ (using (X, W) with explicit intercept). The objective combines: (a) labeled fit of β on (X_L, Y_L) , and (b) unlabeled agreement between γ and β on X_U via the constraint-coupling term. A labeled rich-view fit constraint (budget ν) enforces that γ explains labeled outcomes sufficiently well; the corresponding dual variable is recorded as the Lagrange multiplier λ (used for the main plots). For numerical stability, we add a tiny ridge term 10^{-6} on both β and γ (to prevent solver drift in nearly-flat regions).

Figure-specific settings. (i) Figures 1 : $\text{corr}(X, W) = 0.1$, `seed=1`; and (ii) Figures 1 : $\text{corr}(X, W) = 0.6$, `seed=789`; (iii) Figure 5 plots the test MSE as a function of the Lagrangian parameter λ , where the reported MSE is averaged over 20 random seeds (base `seed = 42`). This figure is the multi-seed, dual-variable analogue of the single-run results shown in Figures 1 and 2.

B.4. PneumoniaMNIST: An Experiment for Algorithm 1

We evaluate Algorithm 1 on a medical image classification task based on PneumoniaMNIST. Each example is a 28×28 grayscale chest image, flattened as deployment features $X \in \mathbb{R}^{784}$. We construct a synthetic privileged proxy $W \in \mathbb{R}^{18}$ available only at training time: the first coordinate is a noisy copy of Y (simulating a strong proxy), and the remaining 17 coordinates are random linear projections of X . This controlled design isolates the algorithm’s ability to distill privileged

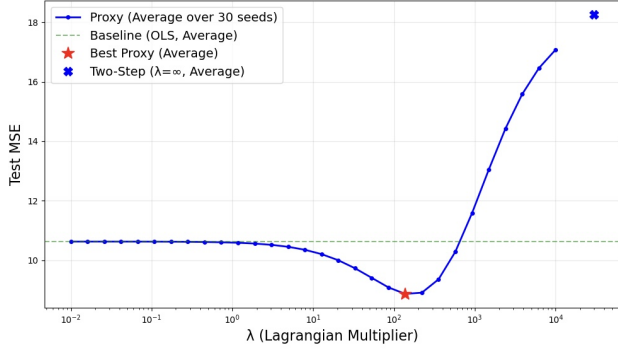


Figure 5. Synthetic Linear Data: U-shaped test MSE vs. λ . Setup: $n = 150$, $m = 6000$, $n_{\text{test}} = 2000$; $d_X = 5$, $d_W = 30$; $\sigma = 1$; $\text{corr}(X, W) = 0.9$. Model: Linear.

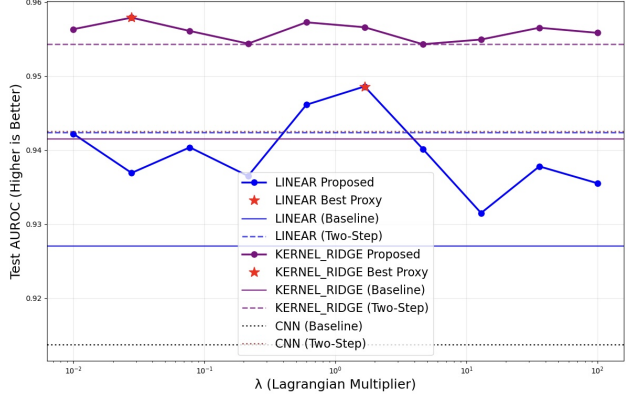


Figure 6. PneumoniaMNIST: Test AUROC vs. λ for Algorithm 1. Setup: $n = 100$, $m = 2000$, $n_{\text{test}} = 1500$; $d_X = 784$, $d_W = 18$; $K = 70$. Model: Linear and Kernel-ridge.

information into a predictor $\hat{f}(X)$. The label $Y \in \{0, 1\}$ indicates pneumonia. We follow the semi-supervised LUPi setup: labeled samples provide (X_i, W_i, Y_i) , while unlabeled samples provide (X_j, W_j) .

Proxy construction (W). Let $X \in \mathbb{R}^{N \times 784}$ be pixel features scaled to $[0, 1]$. We generate $W \in \mathbb{R}^{N \times 18}$ by (i) $W_{:,1} = Y + \varepsilon$ with i.i.d. $\varepsilon \sim \mathcal{N}(0, 0.3^2)$, and (ii) $W_{:,2:18} = XP$ where $P \in \mathbb{R}^{784 \times 17}$ has i.i.d. entries $\mathcal{N}(0, 0.1^2)$ (fixed given the seed).

Splits and preprocessing. We use a single stratified split with $n = 100$ labeled points, $m = 2000$ unlabeled points, and $n_{\text{test}} = 1500$ test points (seed=42). For all linear/kernel methods, we standardize both X and W using `StandardScaler` fit on labeled+unlabeled data only (the test set is never used for fitting). We also feed standardized W to teacher models for stable optimization.

AFS instantiations. We run alternating forward selection in $\mathcal{H}^\star = \mathcal{H}_f \times \mathcal{H}_g$ (Section 3.2) with $K = 70$ iterations, squared loss, and evaluation by test AUROC using the deployment score $f(X)$. We consider:

- **Linear AFS (random-feature dictionary).** Embedded dictionaries use fixed Gaussian random projections with sizes `dict_size_f` = `dict_size_g` = 2048 for f and g respectively.
- **Kernelized AFS (RBF kernel dictionary).** Centers are all labeled points plus up to 500 randomly chosen unlabeled points. Kernel bandwidths γ_f (for X) and γ_g (for (X, W)) are initialized by the median heuristic (computed on up to 600 samples), and tuned by stratified 5-fold CV on labeled data, maximizing AUROC over $\gamma \in \gamma_0 \cdot \{0.25, 0.5, 1, 2, 4\}$ and $\alpha \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$.

λ sweep. We sweep λ over a log-spaced grid of 10 points from 10^{-2} to 10^2 .

Refit with unlabeled pseudo-targets (AFS). After greedy selection, we refit the deployment predictor by ridge regression with $\alpha_{\text{refit}} = 10^{-3}$. When enabled, we obtain pseudo-targets $\tilde{Y}_U = g(X_U, W_U)$ from a teacher g trained on labeled data using $(X, W) \rightarrow Y$ (ridge in the linear case; kernel ridge in the kernel case with tuned α_g), and refit on $(X_L \cup X_U) \rightarrow (Y_L \cup \tilde{Y}_U)$ with sample weights 1 (labeled) and 0.3 (unlabeled).

Baselines and CNN comparisons. **Baseline** trains f on labeled data only using $X \rightarrow Y$ (ridge for linear; kernel ridge with tuned (γ_f, α_f) for kernel). **Two-step** trains a teacher on labeled data using $(X, W) \rightarrow Y$, pseudo-labels unlabeled data, then trains a student on $(X_L \cup X_U) \rightarrow (Y_L \cup \tilde{Y}_U)$ using the same model class. For reference, we also report a CNN baseline (image-only student) and a CNN two-step teacher–student pipeline where the teacher fuses image features with standardized W , pseudo-labels unlabeled data (threshold 0.5), and retrain the image-only student; we use Adam with learning rate 10^{-3} , batch size 32, and 30 epochs.

Metrics. Models output real-valued scores; we apply a sigmoid to obtain probabilities and report test AUROC (primary), along with accuracy at threshold 0.5 and probability MSE against $\{0, 1\}$ targets.

C. Proofs

In what follows, let $a = \frac{n}{N}$, $b = \frac{m}{N}$, and $c = b + a\lambda = \frac{m+n\lambda}{N}$, and define

$$s_f(z) := \frac{b f(x) + a\lambda \eta(z)}{c}, \quad t_g(x) := a \mu(x) + b \mathbb{E}[g(Z) \mid X = x].$$

For functions $f : \mathcal{X} \rightarrow \mathbb{R}$ and $g : \mathcal{Z} \rightarrow \mathbb{R}$, we define the (squared) L^2 norms $\|f\|_X^2 = \mathbb{E}[f(X)^2]$ and $\|g\|_Z^2 = \mathbb{E}[g(Z)^2]$, where the expectations are taken with respect to the marginal distributions of X and Z , respectively. Define the inner products $\langle \cdot, \cdot \rangle_X$ and $\langle \cdot, \cdot \rangle_Z$ analogously.

For fixed points $s^n = (s_1, \dots, s_n)$ in a measurable space $\mathcal{S}$, let $\hat{P}_n$ denote the empirical measure with respect to these points. For a function $h : \mathcal{S} \rightarrow \mathbb{R}$, we write $\|h\|_{L_p(\hat{P}_n)} = \left(\frac{1}{n} \sum_{i=1}^n |h(s_i)|^p\right)^{1/p}$. An L_p - ε -cover of a class $\mathcal{C}$ with respect to $\|\cdot\|_{L_p(\hat{P}_n)}$ is a finite collection $\{h_1, \dots, h_M\} \subset \mathcal{C}$ such that every $h \in \mathcal{C}$ is within ε of some h_j under this norm, and the minimal such M is the empirical covering number $\mathcal{N}_p(\varepsilon, \mathcal{C}, s^n)$. If s^n is an i.i.d. sample, then $\mathcal{N}_p(\varepsilon, \mathcal{C}, S^n)$ becomes a random variable.

C.1. Proof of Theorem 2.1

For a fixed $f \in \mathcal{F}$, let d denote constants that does not depend on g (and may change from line to line). Then, we can write the population loss as follows

$$\begin{aligned} \mathcal{L}(f, g; \lambda) &= b\mathbb{E}[(g(Z) - f(X))^2] + a\lambda\mathbb{E}[(Y - g(Z))^2] + d \\ &= b \left(\mathbb{E}g^2(Z) - 2\mathbb{E}g(Z)f(X) + \mathbb{E}f^2(X) \right) + a\lambda \left(\mathbb{E}Y^2 - 2\mathbb{E}[Yg(Z)] + \mathbb{E}g^2(Z) \right) + d \\ &= c\mathbb{E}g^2(Z) - 2\mathbb{E}[g(Z)(bf(X) + a\lambda Y)] + d \\ &= c\mathbb{E}g^2(Z) - 2\mathbb{E}[g(Z)\mathbb{E}[(bf(X) + a\lambda Y) \mid Z]] + d \\ &= c \left(\mathbb{E}g^2(Z) - 2\mathbb{E}[g(Z)s_f(Z)] \right) + d \\ &= c \left(\mathbb{E}g^2(Z) - 2\mathbb{E}[g(Z)s_f(Z) + \mathbb{E}s_f^2(Z)] \right) + d \\ &= c \|g - s_f\|_Z^2 + d. \end{aligned} \tag{7}$$

Thus, for fixed $f \in \mathcal{F}$, the best response over $\mathcal{G}$ is the metric projection

$$g_{f,\lambda}^{\mathcal{G}} = \operatorname{argmin}_{g \in \mathcal{G}} \mathcal{L}(f, g; \lambda) = \operatorname{argmin}_{g \in \mathcal{G}} \|g - s_f\|_Z^2 = \Pi_{\mathcal{G}} s_f \tag{8}$$

For a fixed $g \in \mathcal{G}$, let d denote constants that does not depend on f (and may change from line to line). Then using same approach that was used to derive (7), we get

$$\begin{aligned} \mathcal{L}(f, g; \lambda) &= a\mathbb{E}[(Y - f(X))^2] + b\mathbb{E}[(g(Z) - f(X))^2] + d \\ &= (a + b)\mathbb{E}f^2(X) - 2\mathbb{E}[f(X)\mathbb{E}[(bg(Z) + aY) \mid X]] + d \\ &= \|f - t_g\|_X^2 + d. \end{aligned} \tag{9}$$

Thus, for fixed $g \in \mathcal{G}$, the best response over $\mathcal{F}$ is

$$f_g^{\mathcal{F}} = \operatorname{argmin}_{f \in \mathcal{F}} \mathcal{L}(f, g; \lambda) = \operatorname{argmin}_{f \in \mathcal{F}} \|f - t_g\|_X^2 = \Pi_{\mathcal{F}} t_g. \tag{10}$$

Hence, the desired result follows from (8) and (10).

Furthermore, if $\mu \in \mathcal{F}$ and $\mu \in \mathcal{G}$ (identifying μ with its lift $\tilde{\mu}(x, w) = \mu(x)$ on $\mathcal{Z}$), and $\eta \in \mathcal{G}$, then

$$\begin{aligned} f^*(x) &= \Pi_{\mathcal{F}} \left(a\mu(x) + b\mathbb{E} \left[\Pi_{\mathcal{G}} \frac{bf^*(X) + a\lambda\eta(Z)}{c} \mid X = x \right] \right) \\ &= \Pi_{\mathcal{F}} \left(a\mu(x) + b\mathbb{E} \left[\frac{b\Pi_{\mathcal{G}}f^*(X) + a\lambda\Pi_{\mathcal{G}}\eta(Z)}{c} \mid X = x \right] \right) \\ &= \Pi_{\mathcal{F}} \left(a\mu(x) + b \left(\frac{b}{c} \mathbb{E} [\Pi_{\mathcal{G}}f^*(X) \mid X = x] + \frac{a\lambda}{c} \mu(x) \right) \right). \end{aligned}$$

It is clear that $f^* = \mu$ satisfies this equation, which means that it is a global minimizer, and thus, $g^* = \frac{m}{m+n\lambda}\mu + \frac{n\lambda}{m+n\lambda}\eta$.

Next, before proving Theorem 2.6, we present a sequence of intermediate results that will be useful for its proof.

Theorem C.1. *The following identity decomposes the excess population loss around (f^*, g^*) :*

$$\begin{aligned} \mathcal{L}(\hat{f}, \hat{g}; \lambda) - \mathcal{L}(f^*, g^*; \lambda) &= \|u\|_X^2 + c\|v\|_Z^2 - 2b\langle u, v \rangle_Z \\ &\quad + \langle \nabla_f \mathcal{L}(f^*, g^*; \lambda), u \rangle_X + \langle \nabla_g \mathcal{L}(f^*, g^*; \lambda), v \rangle_Z, \end{aligned} \quad (11)$$

where

$$u := \hat{f} - f^* \in \mathcal{F} - f^*, \quad v := \hat{g} - g^* \in \mathcal{G} - g^*.$$

Proof. We start by breaking the loss $\mathcal{L}(\hat{f}, \hat{g}; \lambda)$ into three terms as follows

$$\mathcal{L}(\hat{f}, \hat{g}; \lambda) = \textcircled{1} + \textcircled{2} + \textcircled{3},$$

where

$$\textcircled{1} = a\mathbb{E}[(Y - \hat{f}(X))^2], \quad \textcircled{2} = b\mathbb{E}[(\hat{g}(Z) - \hat{f}(X))^2], \quad \text{and} \quad \textcircled{3} = a\lambda\mathbb{E}[(Y - \hat{g}(Z))^2].$$

Next by writing $\hat{f} = f^* + u$, and $\hat{g} = g^* + v$, we can expand each term as done below.

$$\begin{aligned} \textcircled{1} &= a \left(\mathbb{E}[(Y - f^*(X))^2] + \mathbb{E}u^2(X) - 2\mathbb{E}[(Y - f^*(X))u(X)] \right) \\ &= a\mathbb{E}[(Y - f^*(X))^2] + a\|u\|_X^2 - 2\mathbb{E}[u(X)(\mathbb{E}(Y \mid X) - f^*(X))] \\ &= a\mathbb{E}[(Y - f^*(X))^2] + a\|u\|_X^2 - 2a\langle \mu - f^*, u \rangle_X. \\ \textcircled{2} &= b \left(\mathbb{E}[(g^*(Z) - f^*(X))^2] + \mathbb{E}[(v(Z) - u(X))^2] + 2\mathbb{E}[(g^*(Z) - f^*(X))(v(Z) - u(X))] \right) \\ &= b\mathbb{E}[(g^*(Z) - f^*(X))^2] + b\|v\|_Z^2 + b\|u\|_X^2 - 2b\langle u, v \rangle_Z \\ &\quad + 2b\langle g^* - f^*, v \rangle_Z - 2b\langle \tilde{g}^* - f^*, u \rangle_X, \end{aligned}$$

where $\tilde{g}^*(x) = \mathbb{E}[g^*(Z) \mid X = x]$.

$$\begin{aligned} \textcircled{3} &= a\lambda\mathbb{E}[(Y - g^*(Z))^2] + a\lambda\|v\|_Z^2 - 2a\lambda\mathbb{E}[(Y - g^*(Z))v(Z)] \\ &= a\lambda\mathbb{E}[(Y - g^*(Z))^2] + a\lambda\|v\|_Z^2 - 2a\lambda\langle \eta - g^*, v \rangle_Z. \end{aligned}$$

Based on the above expansions, observe that

$$\begin{aligned} \mathcal{L}(\hat{f}, \hat{g}; \lambda) - \mathcal{L}(f^*, g^*; \lambda) &= (a+b)\|u\|_X^2 + c\|v\|_Z^2 - 2b\langle u, v \rangle_Z \\ &\quad + \langle 2(f^* - b\tilde{g}^* - a\mu), u \rangle_X + \langle 2(cg^* - bf^* - a\lambda\eta), v \rangle_Z \\ &= \|u\|_X^2 + c\|v\|_Z^2 - 2b\langle u, v \rangle_Z \\ &\quad + \langle 2(f^* - t_{g^*}), u \rangle_X + \langle 2c(g^* - s_{f^*}), v \rangle_Z \\ &= \|u\|_X^2 + c\|v\|_Z^2 - 2b\langle u, v \rangle_Z \\ &\quad + \langle \nabla_f \mathcal{L}(f^*, g^*; \lambda), u \rangle_X + \langle \nabla_g \mathcal{L}(f^*, g^*; \lambda), v \rangle_Z, \end{aligned}$$

where the last equality follows using (7) and (9) which yield that

$$\nabla_g \mathcal{L}(f, g; \lambda) = 2c(g - s_f), \quad \nabla_f \mathcal{L}(f, g; \lambda) = 2(f - t_g).$$

□

C.2. Proof of Corollary 2.3

Since (f^*, g^*) minimizes $\mathcal{L}$ over the closed convex set $\mathcal{F} \times \mathcal{G}$, the variational inequalities give

$$\langle \nabla_f \mathcal{L}(f^*, g^*; \lambda), u \rangle_X \geq 0, \quad \langle \nabla_g \mathcal{L}(f^*, g^*; \lambda), v \rangle_Z \geq 0$$

for all $u \in \mathcal{F} - f^*, v \in \mathcal{G} - g^*$.

Thus, dropping the nonnegative linear terms in (11) yields

$$\mathcal{L}(\hat{f}, \hat{g}; \lambda) - \mathcal{L}(f^*, g^*; \lambda) \geq \|u\|_X^2 + c\|v\|_Z^2 - 2b\langle u, v \rangle_Z. \quad (12)$$

The result follows by taking expectation with respect to D of both sides (12), then we can further lower bound the right hand by completing the square of the last two terms as done below.

$$\begin{aligned} & \mathbb{E}_{\mathcal{D}} \|u\|_X^2 + c\mathbb{E}_{\mathcal{D}} \|v\|_Z^2 - 2b\mathbb{E}_{\mathcal{D}} \langle u, v \rangle_Z \\ &= \mathbb{E}_{\mathcal{D}} \|u\|_X^2 + c\mathbb{E}_{\mathcal{D}} \|v\|_Z^2 - 2b\rho_{\star} \sqrt{\mathbb{E}_{\mathcal{D}} \|u\|_X^2 \mathbb{E}_{\mathcal{D}} \|v\|_Z^2} \\ &= \mathbb{E}_{\mathcal{D}} \|u\|_X^2 + c \left(\mathbb{E}_{\mathcal{D}} \|v\|_Z^2 - \frac{2b\rho_{\star}}{c} \sqrt{\mathbb{E}_{\mathcal{D}} \|u\|_X^2 \mathbb{E}_{\mathcal{D}} \|v\|_Z^2} + \frac{b^2\rho_{\star}^2}{c^2} \mathbb{E}_{\mathcal{D}} \|u\|_X^2 \right) - \frac{b^2\rho_{\star}^2}{c} \mathbb{E}_{\mathcal{D}} \|u\|_X^2 \\ &= \gamma_{n,m,\lambda}(\rho_{\star}) \mathbb{E}_{\mathcal{D}} \|u\|_X^2 + c \left(\sqrt{\mathbb{E}_{\mathcal{D}} \|v\|_Z^2} - \frac{b\rho_{\star}}{c} \sqrt{\mathbb{E}_{\mathcal{D}} \|u\|_X^2} \right)^2 \\ &\geq \gamma_{n,m,\lambda}(\rho_{\star}) \mathbb{E}_{\mathcal{D}} \|u\|_X^2. \end{aligned}$$

For notational convenience, throughout the remainder of the text we write $\sum_i$ for $\sum_{i=1}^n$ and $\sum_j$ for $\sum_{j=n+1}^N$.

Define

$$\begin{aligned} k_i(f, g) &:= (f(X_i) - Y_i)^2 + \lambda(g(Z_i) - Y_i)^2 - [(f^*(X_i) - Y_i)^2 + \lambda(g^*(Z_i) - Y_i)^2], \\ p_j(f, g) &:= (f(X_j) - g(Z_j))^2 - (f^*(X_j) - g^*(Z_j))^2. \end{aligned}$$

We now state a result adapted from Györfi et al. (2002, Theorem 11.4).

Theorem C.2. Assume that 2.5 holds.

Then

$$\begin{aligned} & \mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \mathbb{E}[\hat{\mathcal{L}}(f, g; \lambda)] - \mathbb{E}[\hat{\mathcal{L}}(f^*, g^*; \lambda)] - \frac{1}{N} \left(\sum_i k_i + \sum_j p_j \right) \right. \\ & \quad \left. \geq \epsilon \left(\alpha + \beta + \mathbb{E}[\hat{\mathcal{L}}(f, g; \lambda)] - \mathbb{E}[\hat{\mathcal{L}}(f^*, g^*; \lambda)] \right) \right\} \\ & \leq 10\mathfrak{M}_{\mathcal{F}, \mathcal{G}}^{n,m,\lambda} \exp \left(-\frac{\epsilon^2(1-\epsilon)\alpha N}{512B^2(\lambda+1)} \right), \end{aligned}$$

where $\alpha, \beta > 0$, $0 < \epsilon < 1$, and

$$\begin{aligned} \mathfrak{M}_{\mathcal{F}, \mathcal{G}}^{n, m, \lambda} := & \max_{Z^n} \mathcal{N}_1 \left(\frac{(\alpha + \beta)}{80B^2\sqrt{\lambda+1}a}, \mathcal{F}, X^n \right) \cdot \mathcal{N}_1 \left(\frac{(\alpha + \beta)}{80B^2a(\lambda+1)^{3/2}}, \mathcal{G}, Z^n \right) + \\ & \max_{Z^m} \mathcal{N}_1 \left(\frac{(\alpha + \beta)}{80B^2\sqrt{\lambda+1}b}, \mathcal{F}, X^m \right) \cdot \mathcal{N}_1 \left(\frac{(\alpha + \beta)}{80B^2\sqrt{\lambda+1}b}, \mathcal{G}, Z^m \right) + \\ & \max_{Z^N} \mathcal{N}_1 \left(\frac{\epsilon\beta}{40B\sqrt{\lambda+1}}, \mathcal{F}, X^N \right) \cdot \mathcal{N}_1 \left(\frac{\epsilon\beta}{40(\lambda+1)^{3/2}B}, \mathcal{G}, Z^N \right). \end{aligned}$$

Proof. Throughout this proof, for $(f, g) \in \mathcal{F} \times \mathcal{G}$, we interpret f and g pointwise, writing $f \equiv f(X)$, $g \equiv g(Z)$, so that expressions such as $(f - Y)^2$, $(g - Y)^2$, and $(f - g)^2$ are understood as $(f(X) - Y)^2$, $(g(Z) - Y)^2$, and $(f(X) - g(Z))^2$, respectively.

For a sample (X, Z, Y) , define the $*$ -norm

$$\|(f, g) - (Y, Y)\|_*^2 := a(f - Y)^2 + a\lambda(g - Y)^2 + b(f - g)^2.$$

Denote

$$A_{f, g} := \|(f, g) - (Y, Y)\|_*^2 - \|(f^*, g^*) - (Y, Y)\|_*^2 = a k(f, g) + b p(f, g),$$

where $k(f, g) = (f - Y)^2 + \lambda(g - Y)^2 - [(f^* - Y)^2 + \lambda(g^* - Y)^2]$, and $p(f, g) = (f - g)^2 - (f^* - g^*)^2$.

Based on the above definition, our goal is to bound

$$\mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \mathbb{E}[A_{f, g}] - \frac{1}{N} \sum_i k_i - \frac{1}{N} \sum_j p_j \geq \epsilon(\alpha + \beta + \mathbb{E}[A_{f, g}]) \right\} \quad (13)$$

Observe that when $N \leq \frac{128(\lambda+1)B^2}{\epsilon^2(\alpha+\beta)}$, we have

$$10\mathfrak{M}_{\mathcal{F}, \mathcal{G}}^{n, m, \lambda} \exp \left(-\frac{\epsilon^2(1-\epsilon)\alpha N}{512B^2(\lambda+1)} \right) \geq 15 \exp \left(-\frac{\epsilon^2(1-\epsilon)(\alpha+\beta)N}{512B^2(\lambda+1)} \right) \geq 15 \exp \left(-\frac{128}{512} \right) > 1,$$

thus it is sufficient to only consider the case when $N > \frac{128(\lambda+1)B^2}{\epsilon^2(\alpha+\beta)}$.

In order to bound (13) we will break the proof in multiple steps as follows

Step 1: Symmetrization

Let $\mathcal{D}'_L = \{(X'_i, W'_i, Y'_i)\}_{i=1}^n \stackrel{\text{i.i.d.}}{\sim} P_{x, w, y}$, and $\mathcal{D}'_U = \{Z'_j = (X'_j, W'_j)\}_{j=n+1}^N \stackrel{\text{i.i.d.}}{\sim} P_{x, w}$, be a *ghost* sample independent of $\mathcal{D}_L$ and $\mathcal{D}_U$, respectively. Denote

$$\begin{aligned} k'_i &= (f(X'_i) - Y'_i)^2 + \lambda(g(Z'_i) - Y'_i)^2 - (f^*(X'_i) - Y'_i)^2 - \lambda(g^*(Z'_i) - Y'_i)^2, \\ p'_j &= (f(X'_j) - g(Z'_j))^2 - (f^*(X'_j) - g^*(Z'_j))^2, \end{aligned}$$

and define the two independent and identically distributed sets

$$\mathcal{D} = \{k_i, p_j : 1 \leq i \leq n, n+1 \leq j \leq N\}, \quad \mathcal{D}' = \{k'_i, p'_j : 1 \leq i \leq n, n+1 \leq j \leq N\}.$$

We will show that

$$\begin{aligned} & \mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \mathbb{E}[A_{f, g}] - \frac{1}{N} \sum_i k_i - \frac{1}{N} \sum_j p_j \geq \epsilon(\alpha + \beta + \mathbb{E}[A_{f, g}]) \right\} \\ & \leq \frac{8}{7} \mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \frac{1}{N} \left(\sum_i (k'_i - k_i) + \sum_j (p'_j - p_j) \right) \geq \frac{\epsilon}{2} (\alpha + \beta + \mathbb{E}[A_{f, g}]) \right\} \quad (14) \end{aligned}$$

For that matter, consider

$$\begin{aligned}
 \text{Var}\left(\frac{1}{N}\left(\sum_i k'_i + \sum_j p'_j\right)\right) &= \frac{1}{N^2}\text{Var}\left(\sum_i k'_i + \sum_j p'_j\right) \leq \frac{1}{N^2}\left(n\text{Var}(k'_1) + m\text{Var}(p'_1)\right) \\
 &\leq \frac{a}{N}\mathbb{E}\left[\left((f-y)^2 + \lambda(g-y)^2 - (f^\star-y)^2 - \lambda(g^\star-y)^2\right)^2\right] + \frac{b}{N}\mathbb{E}\left[\left((f-g)^2 - (f^\star-g^\star)^2\right)^2\right] \\
 &\leq \frac{1}{N}\mathbb{E}\left[a((f+f^\star-2y)(f-f^\star) + \lambda(g+g^\star-2y)(g-g^\star))^2\right. \\
 &\quad \left.+ b((f-g+f^\star-g^\star)(f-g-f^\star+g^\star))^2\right] \\
 &\stackrel{(*)}{\leq} \frac{1}{N}\mathbb{E}\left[a((f+f^\star-2y)^2 + \lambda(g+g^\star-2y)^2)((f-f^\star)^2 + \lambda(g-g^\star)^2)\right. \\
 &\quad \left.+ b((f-g+f^\star-g^\star)(f-g-f^\star+g^\star))^2\right] \\
 &\leq \frac{16(\lambda+1)B^2}{N}\mathbb{E}\left[(a((f-f^\star)^2 + \lambda(g-g^\star)^2) + b(f-f^\star+g^\star-g)^2)\right] \\
 &= \frac{16(\lambda+1)B^2}{N}\mathbb{E}\left[\|(f,g) - (f^\star,g^\star)\|_*^2\right] \\
 &\stackrel{(\dagger)}{\leq} \frac{16(\lambda+1)B^2}{N}\mathbb{E}\left[\|(f,g) - (Y,Y)\|_*^2 - \|(f^\star,g^\star) - (Y,Y)\|_*^2\right] \\
 &\leq \frac{16(\lambda+1)B^2}{N}\mathbb{E}\left[A_{f,g}\right], \tag{15}
 \end{aligned}$$

where $(*)$ follows by applying Cauchy–Schwarz, and $(\dagger)$ holds since $(f^\star, g^\star)$ minimizes $\mathbb{E}\|(f,g) - (Y,Y)\|_*^2 = \mathcal{L}(f,g;\lambda)$.

Thus, we have

$$\mathbb{E}\left[A_{f,g}\right] \geq \frac{1}{16(\lambda+1)B^2}\left(a\mathbb{E}[k^2(f,g)] + b\mathbb{E}[p^2(f,g)]\right) \tag{16}$$

Next, find $f_{\mathcal{D}}, g_{\mathcal{D}}$ such that

$$\mathbb{E}\left[A_{f_{\mathcal{D}},g_{\mathcal{D}}} \mid \mathcal{D}\right] - \frac{1}{N}\sum_i k_i(f_{\mathcal{D}},g_{\mathcal{D}}) - \frac{1}{N}\sum_j p_j(f_{\mathcal{D}},g_{\mathcal{D}}) \geq \epsilon\left(\alpha + \beta + \mathbb{E}\left[A_{f_{\mathcal{D}},g_{\mathcal{D}}} \mid \mathcal{D}\right]\right).$$

And consider

$$\begin{aligned}
 &\mathbb{P}\left\{\mathbb{E}\left[A_{f_{\mathcal{D}},g_{\mathcal{D}}} \mid \mathcal{D}\right] - \frac{1}{N}\sum_i k'_i(f_{\mathcal{D}},g_{\mathcal{D}}) - \frac{1}{N}\sum_j p'_j(f_{\mathcal{D}},g_{\mathcal{D}}) \geq \frac{\epsilon}{2}\left(\alpha + \beta + \mathbb{E}\left[A_{f_{\mathcal{D}},g_{\mathcal{D}}} \mid \mathcal{D}\right]\right)\right\} \\
 &\leq \frac{\text{Var}\left(\frac{1}{N}\left(\sum_i k'_i + \sum_j p'_j\right) \mid \mathcal{D}\right)}{\left(\frac{\epsilon}{2}\left(\alpha + \beta + \mathbb{E}\left[A_{f_{\mathcal{D}},g_{\mathcal{D}}} \mid \mathcal{D}\right]\right)\right)^2} \\
 &\stackrel{(*)}{\leq} \frac{64(\lambda+1)B^2}{N\epsilon^2\left(\alpha + \beta + \mathbb{E}\left[A_{f_{\mathcal{D}},g_{\mathcal{D}}} \mid \mathcal{D}\right]\right)^2}\mathbb{E}\left[A_{f_{\mathcal{D}},g_{\mathcal{D}}} \mid \mathcal{D}\right] \\
 &\leq \frac{16(\lambda+1)B^2}{N\epsilon^2(\alpha + \beta)} < \frac{1}{8},
 \end{aligned}$$

where $(*)$ follows by (15) and in the last inequality we used $N > \frac{128(\lambda+1)B^2}{\epsilon^2(\alpha + \beta)}$.

Define the two events

$$\begin{aligned}\mathcal{A} &:= \left\{ \mathbb{E}[A_{f_{\mathcal{D}}, g_{\mathcal{D}}} | \mathcal{D}] - \frac{1}{N} \sum_i k_i(f_{\mathcal{D}}, g_{\mathcal{D}}) - \frac{1}{N} \sum_j p_j(f_{\mathcal{D}}, g_{\mathcal{D}}) \geq \epsilon \left(\alpha + \beta + \mathbb{E}[A_{f_{\mathcal{D}}, g_{\mathcal{D}}} | \mathcal{D}] \right) \right\} \\ \mathcal{B} &:= \left\{ \mathbb{E}[A_{f_{\mathcal{D}}, g_{\mathcal{D}}} | \mathcal{D}] - \frac{1}{N} \sum_i k'_i(f_{\mathcal{D}}, g_{\mathcal{D}}) - \frac{1}{N} \sum_j p'_j(f_{\mathcal{D}}, g_{\mathcal{D}}) \leq \frac{\epsilon}{2} \left(\alpha + \beta + \mathbb{E}[A_{f_{\mathcal{D}}, g_{\mathcal{D}}} | \mathcal{D}] \right) \right\}\end{aligned}$$

Hence,

$$\begin{aligned}& \mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \mathbb{E}[A_{f, g}] - \frac{1}{N} \sum_i k_i - \frac{1}{N} \sum_j p_j \geq \epsilon \left(\alpha + \beta + \mathbb{E}[A_{f, g}] \right) \right\} \\ &= \mathbb{P} \left\{ \mathbb{E}[A_{f_{\mathcal{D}}, g_{\mathcal{D}}} | \mathcal{D}] - \frac{1}{N} \sum_i k_i(f_{\mathcal{D}}, g_{\mathcal{D}}) - \frac{1}{N} \sum_j p_j(f_{\mathcal{D}}, g_{\mathcal{D}}) \geq \epsilon \left(\alpha + \beta + \mathbb{E}[A_{f_{\mathcal{D}}, g_{\mathcal{D}}} | \mathcal{D}] \right) \right\} \\ &= \mathbb{P}\{\mathcal{A} \cap \mathcal{B}\} / \mathbb{P}\{\mathcal{B}\} \leq \frac{8}{7} \mathbb{P}\{\mathcal{A} \cap \mathcal{B}\} \\ &\leq \frac{8}{7} \mathbb{P} \left\{ \frac{1}{N} \sum_i k'_i(f_{\mathcal{D}}, g_{\mathcal{D}}) + \sum_j p'_j(f_{\mathcal{D}}, g_{\mathcal{D}}) - \sum_i k_i(f_{\mathcal{D}}, g_{\mathcal{D}}) - \sum_j p_j(f_{\mathcal{D}}, g_{\mathcal{D}}) \geq \frac{\epsilon}{2} \left(\alpha + \beta + \mathbb{E}[A_{f_{\mathcal{D}}, g_{\mathcal{D}}} | \mathcal{D}] \right) \right\} \\ &\leq \frac{8}{7} \mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \frac{1}{N} \left(\sum_i k'_i(f, g) - k_i(f, g) + \sum_j p'_j(f, g) - p_j(f, g) \right) \geq \frac{\epsilon}{2} \left(\alpha + \beta + \mathbb{E}[A_{f, g}] \right) \right\}.\end{aligned}$$

Therefore (14) holds.

Step 2: Population to empirical

Next we will show that

$$\begin{aligned}& \mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \frac{1}{N} \left(\sum_i (k'_i - k_i) + \sum_j (p'_j - p_j) \right) \geq \frac{\epsilon}{2} \left(\alpha + \beta + \mathbb{E}[A_{f, g}] \right) \right\} \\ &\leq \mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \frac{1}{N} \left(\sum_i (k'_i - k_i) + \sum_j (p'_j - p_j) \right) \geq \frac{\epsilon}{2} (\alpha + \beta) \right. \\ &\quad \left. + \frac{\epsilon}{64(\lambda + 1)B^2} \frac{1 - \epsilon}{1 + \epsilon} \left(\frac{1}{N} \left(\sum_i (k_i^2 + k_i'^2) + \sum_j (p_j^2 + p_j'^2) \right) - \frac{2\epsilon}{1 - \epsilon} (\alpha + \beta) \right) \right\} \\ &\quad + 8\mathbb{E} \left[\mathcal{N}_1 \left(\frac{\alpha + \beta}{10a}, \mathcal{G}_k, Z^n \right) + \mathcal{N}_1 \left(\frac{\alpha + \beta}{10b}, \mathcal{G}_p, Z^m \right) \right] \exp \left(-\frac{\epsilon^2(\alpha + \beta)N}{480B^4(\lambda + 1)^2} \right).\end{aligned}\tag{17}$$

In order to prove (17), we first present the following lemma.

Lemma C.3. Let $\mathcal{G}_k = \{k(f, g) : (f, g) \in \mathcal{F} \times \mathcal{G}\}$, $\mathcal{G}_p = \{p(f, g) : (f, g) \in \mathcal{F} \times \mathcal{G}\}$. Z^n, Z^m are the samples used to generate k_i, p_j respectively. Then

$$\begin{aligned}& \mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \frac{\frac{1}{N} \left(\sum_i k_i^2 + \sum_j p_j^2 \right) - \mathbb{E}[ak^2 + bp^2]}{\alpha + \beta + \frac{1}{N} \left(\sum_i k_i^2 + \sum_j p_j^2 \right) + \mathbb{E}[ak^2 + bp^2]} \geq \epsilon \right\} \\ &\leq 4\mathbb{E} \left[\mathcal{N}_1 \left(\frac{\alpha + \beta}{10a}, \mathcal{G}_k, Z^n \right) + \mathcal{N}_1 \left(\frac{\alpha + \beta}{10b}, \mathcal{G}_p, Z^m \right) \right] \exp \left(-\frac{\epsilon^2(\alpha + \beta)N}{480B^4(\lambda + 1)^2} \right).\end{aligned}$$

Proof. The desired bound can be obtained as follows

$$\begin{aligned}
 & \mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \frac{\frac{1}{N} \left(\sum_i k_i^2 + \sum_j p_j^2 \right) - \mathbb{E}[ak^2 + bp^2]}{\alpha + \beta + \frac{1}{N} \left(\sum_i k_i^2 + \sum_j p_j^2 \right) + \mathbb{E}[ak^2 + bp^2]} \geq \epsilon \right\} \\
 &= \mathbb{P} \left\{ \sup_{(f, g) \in \mathcal{F} \times \mathcal{G}} \frac{\frac{1}{N} \left(\sum_i k_i^2 + \sum_j p_j^2 \right) - \mathbb{E}[ak^2 + bp^2]}{\alpha + \beta + \frac{1}{N} \left(\sum_i k_i^2 + \sum_j p_j^2 \right) + \mathbb{E}[ak^2 + bp^2]} \geq \epsilon \right\} \\
 &= \mathbb{P} \left\{ \sup_{(f, g) \in \mathcal{F} \times \mathcal{G}} \frac{a \left(\frac{1}{n} \sum_i k_i^2 - \mathbb{E}[k^2] \right) + b \left(\frac{1}{m} \sum_j p_j^2 - \mathbb{E}[p^2] \right)}{\alpha + \beta + a \left(\frac{1}{n} \sum_i k_i^2 + \mathbb{E}[k^2] \right) + b \left(\frac{1}{m} \sum_j p_j^2 + \mathbb{E}[p^2] \right)} \geq \epsilon \right\} \\
 &\leq \mathbb{P} \left\{ \sup_{(f, g) \in \mathcal{F} \times \mathcal{G}} \frac{a \left(\frac{1}{n} \sum_i k_i^2 - \mathbb{E}[k^2] \right)}{\frac{\alpha + \beta}{2} + a \left(\frac{1}{n} \sum_i k_i^2 + \mathbb{E}[k^2] \right)} \geq \epsilon \right\} + \mathbb{P} \left\{ \sup_{(f, g) \in \mathcal{F} \times \mathcal{G}} \frac{b \left(\frac{1}{m} \sum_j p_j^2 - \mathbb{E}[p^2] \right)}{\frac{\alpha + \beta}{2} + b \left(\frac{1}{m} \sum_j p_j^2 + \mathbb{E}[p^2] \right)} \geq \epsilon \right\} \\
 &\leq 4\mathbb{E} \left[\mathcal{N}_1 \left(\frac{\alpha + \beta}{10a}, \mathcal{G}_k, Z^n \right) + \mathcal{N}_1 \left(\frac{\alpha + \beta}{10b}, \mathcal{G}_p, Z^m \right) \right] \exp \left(-\frac{\epsilon^2(\alpha + \beta)N}{480B^4(\lambda + 1)^2} \right),
 \end{aligned}$$

where the last inequality uses Györfi et al. (2002, Theorem 11.6). $\square$

Next, define the events

$$\begin{aligned}
 \mathcal{S}_1 &:= \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \frac{1}{N} \left(\sum_i (k'_i - k_i) + \sum_j (p'_j - p_j) \right) \geq \frac{\epsilon}{2} (\alpha + \beta + \mathbb{E}[A_{f,g}]) \right\}, \\
 \mathcal{S}_2 &:= \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \frac{1}{N} \left(\sum_i k_i^2 + \sum_j p_j^2 \right) - \mathbb{E}[ak^2 + bp^2] \leq \epsilon \left(\alpha + \beta + \frac{1}{N} \left(\sum_i k_i^2 + \sum_j p_j^2 \right) + \mathbb{E}[ak^2 + bp^2] \right) \right\}, \\
 \mathcal{S}_3 &:= \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \frac{1}{N} \left(\sum_i (k'_i)^2 + \sum_j (p'_j)^2 \right) - \mathbb{E}[ak^2 + bp^2] \leq \epsilon \left(\alpha + \beta + \frac{1}{N} \left(\sum_i (k'_i)^2 + \sum_j (p'_j)^2 \right) + \mathbb{E}[ak^2 + bp^2] \right) \right\}.
 \end{aligned}$$

Then we have the set inclusion $\mathcal{S}_1 \subset (\mathcal{S}_1 \cap \mathcal{S}_2 \cap \mathcal{S}_3) \cup \mathcal{S}_2^c \cup \mathcal{S}_3^c$, hence

$$\begin{aligned}
 \mathbb{P}(\mathcal{S}_1) &\leq \mathbb{P}(\mathcal{S}_1 \cap \mathcal{S}_2 \cap \mathcal{S}_3) + \mathbb{P}(\mathcal{S}_2^c) + \mathbb{P}(\mathcal{S}_3^c) \\
 &\leq \mathbb{P}(\mathcal{S}_1 \cap \mathcal{S}_2 \cap \mathcal{S}_3) + 2\mathbb{P}(\mathcal{S}_2^c) \\
 &\leq \mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \frac{1}{N} \left(\sum_i (k'_i - k_i) + \sum_j (p'_j - p_j) \right) \geq \frac{\epsilon}{2} (\alpha + \beta) \right. \\
 &\quad \left. + \frac{\epsilon}{64(\lambda + 1)B^2} \frac{1 - \epsilon}{1 + \epsilon} \left(\frac{1}{N} \left(\sum_i (k_i^2 + k_i'^2) + \sum_j (p_j^2 + p_j'^2) \right) - \frac{2\epsilon}{1 - \epsilon} (\alpha + \beta) \right) \right\} \\
 &\quad + 8\mathbb{E} \left[\mathcal{N}_1 \left(\frac{\alpha + \beta}{10a}, \mathcal{G}_k, Z^n \right) + \mathcal{N}_1 \left(\frac{\alpha + \beta}{10b}, \mathcal{G}_p, Z^m \right) \right] \exp \left(-\frac{\epsilon^2(\alpha + \beta)N}{480B^4(\lambda + 1)^2} \right),
 \end{aligned}$$

where in the last inequality we used Lemma C.3, and using that $\mathcal{S}_2$ implies

$$\mathbb{E}[A_{f,g}] \stackrel{(*)}{\geq} \frac{1}{16(\lambda + 1)B^2} \mathbb{E}[ak^2 + bp^2] \geq \frac{1}{16(\lambda + 1)B^2} \frac{1 - \epsilon}{1 + \epsilon} \left(\frac{1}{N} \left(\sum_i k_i^2 + \sum_j p_j^2 \right) - \frac{\epsilon}{1 - \epsilon} (\alpha + \beta) \right)$$

Similarly for $\mathcal{D}'$, $\mathcal{S}_3$ implies

$$\mathbb{E}[A_{f,g}] \stackrel{(*)}{\geq} \frac{1}{16(\lambda+1)B^2} \mathbb{E}[ak^2 + bp^2] \geq \frac{1}{16(\lambda+1)B^2} \frac{1-\epsilon}{1+\epsilon} \left(\frac{1}{N} \left(\sum_i k_i'^2 + \sum_j p_j'^2 \right) - \frac{\epsilon}{1-\epsilon} (\alpha + \beta) \right),$$

where $(*)$ follows by (16). Therefore (17) holds.

Step 3: Rademacher Variables: We introduce independent Rademacher variables $\{U_i\}_{i=1}^n$ and $\{U'_j\}_{j=n+1}^N$, i.e., i.i.d. random variables satisfying $\mathbb{P}(U_i = 1) = \mathbb{P}(U_i = -1) = \mathbb{P}(U'_j = 1) = \mathbb{P}(U'_j = -1) = \frac{1}{2}$. Then we have

$$\begin{aligned} & \mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \frac{1}{N} \left(\sum_i (k'_i - k_i) + \sum_j (p'_j - p_j) \right) \geq \frac{\epsilon}{2} (\alpha + \beta) \right. \\ & \quad \left. + \frac{\epsilon}{64(\lambda+1)B^2} \frac{1-\epsilon}{1+\epsilon} \left(\frac{1}{N} \left(\sum_i (k_i^2 + k_i'^2) + \sum_j (p_j^2 + p_j'^2) \right) - \frac{2\epsilon}{1-\epsilon} (\alpha + \beta) \right) \right\} \\ & \leq 2\mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \frac{1}{N} \left| \sum_i U_i k_i + \sum_j U_j p_j \right| \geq \frac{\epsilon}{4} (\alpha + \beta) \right. \\ & \quad \left. + \frac{\epsilon}{64(\lambda+1)B^2} \frac{1-\epsilon}{1+\epsilon} \left(\frac{1}{N} \left(\sum_i k_i^2 + \sum_j p_j^2 \right) - \frac{\epsilon}{1-\epsilon} (\alpha + \beta) \right) \right\}, \end{aligned} \quad (18)$$

which can be shown by symmetrization and union bound as follows

$$\begin{aligned} & \mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \frac{1}{N} \left(\sum_i (k'_i - k_i) + \sum_j (p'_j - p_j) \right) \geq \frac{\epsilon}{2} (\alpha + \beta) \right. \\ & \quad \left. + \frac{\epsilon}{64(\lambda+1)B^2} \frac{1-\epsilon}{1+\epsilon} \left(\frac{1}{N} \left(\sum_i (k_i^2 + k_i'^2) + \sum_j (p_j^2 + p_j'^2) \right) - \frac{2\epsilon}{1-\epsilon} (\alpha + \beta) \right) \right\} \\ & = \mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \frac{1}{N} \left(\sum_i U'_i k'_i - U_i k_i + \sum_j U'_j p'_j - U_j p_j \right) \geq \frac{\epsilon}{2} (\alpha + \beta) \right. \\ & \quad \left. + \frac{\epsilon}{64(\lambda+1)B^2} \frac{1-\epsilon}{1+\epsilon} \left(\frac{1}{N} \left(\sum_i (k_i^2 + k_i'^2) + \sum_j (p_j^2 + p_j'^2) \right) - \frac{2\epsilon}{1-\epsilon} (\alpha + \beta) \right) \right\} \\ & \leq \mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \frac{1}{N} \left| \sum_i U_i k_i + \sum_j U_j p_j \right| \geq \frac{\epsilon}{4} (\alpha + \beta) \right. \\ & \quad \left. + \frac{\epsilon}{64(\lambda+1)B^2} \frac{1-\epsilon}{1+\epsilon} \left(\frac{1}{N} \left(\sum_i k_i^2 + \sum_j p_j^2 \right) - \frac{\epsilon}{1-\epsilon} (\alpha + \beta) \right) \right\} \\ & + \mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \frac{1}{N} \left| \sum_i U'_i k'_i + \sum_j U'_j p'_j \right| \geq \frac{\epsilon}{4} (\alpha + \beta) \right. \\ & \quad \left. + \frac{\epsilon}{64(\lambda+1)B^2} \frac{1-\epsilon}{1+\epsilon} \left(\frac{1}{N} \left(\sum_i k_i'^2 + \sum_j p_j'^2 \right) - \frac{\epsilon}{1-\epsilon} (\alpha + \beta) \right) \right\} \\ & = 2\mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \frac{1}{N} \left| \sum_i U_i k_i + \sum_j U_j p_j \right| \geq \frac{\epsilon}{4} (\alpha + \beta) \right. \\ & \quad \left. + \frac{\epsilon}{64(\lambda+1)B^2} \frac{1-\epsilon}{1+\epsilon} \left(\frac{1}{N} \left(\sum_i k_i^2 + \sum_j p_j^2 \right) - \frac{\epsilon}{1-\epsilon} (\alpha + \beta) \right) \right\}. \end{aligned}$$

Therefore (18) holds.

Step 4: Covering Numbers: Define Γ_δ as a smallest subset of $\mathcal{F} \times \mathcal{G}$ such that, for every $(f, g) \in \mathcal{F} \times \mathcal{G}$, there exists $(\tilde{f}, \tilde{g}) \in \Gamma_\delta$ satisfying

$$\frac{1}{N} \left(\sum_i |k_i(f, g) - k_i(\tilde{f}, \tilde{g})| + \sum_j |p_j(f, g) - p_j(\tilde{f}, \tilde{g})| \right) \leq \delta.$$

For notational simplicity, we write $\tilde{k}_i$ and $\tilde{p}_j$ in place of $k_i(\tilde{f}, \tilde{g})$ and $p_j(\tilde{f}, \tilde{g})$, respectively. Then we will show that

$$\begin{aligned} & \mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \frac{1}{N} \left| \sum_i U_i k_i + \sum_j U_j p_j \right| \geq \frac{\epsilon}{4} (\alpha + \beta) + \frac{\epsilon}{64(\lambda+1)B^2} \frac{1-\epsilon}{1+\epsilon} \left(\frac{1}{N} \left(\sum_i k_i^2 + \sum_j p_j^2 \right) - \frac{\epsilon}{1-\epsilon} (\alpha + \beta) \right) \right\} \\ & \leq |\Gamma_{\frac{\epsilon\beta}{5}}| \max_{(\tilde{f}, \tilde{g}) \in \Gamma_{\frac{\epsilon\beta}{5}}} \mathbb{P} \left\{ \frac{1}{N} \left| \sum_i U_i \tilde{k}_i + \sum_j U_j \tilde{p}_j \right| \geq \frac{\epsilon}{4} \alpha + \frac{\epsilon}{64(\lambda+1)B^2} \frac{1-\epsilon}{1+\epsilon} \left(\frac{1}{N} \left(\sum_i \tilde{k}_i^2 + \sum_j \tilde{p}_j^2 \right) - \frac{\epsilon}{1-\epsilon} \alpha \right) \right\}. \end{aligned} \quad (19)$$

We show (19) as follows. Conditioned on $\mathcal{D}$, we have

$$\begin{aligned} & \mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \frac{1}{N} \left| \sum_i U_i k_i + \sum_j U_j p_j \right| \geq \frac{\epsilon}{4} (\alpha + \beta) + \frac{\epsilon}{64(\lambda+1)B^2} \frac{1-\epsilon}{1+\epsilon} \left(\frac{1}{N} \left(\sum_i k_i^2 + \sum_j p_j^2 \right) - \frac{\epsilon}{1-\epsilon} (\alpha + \beta) \right) \right\} \\ & \stackrel{(*)}{\leq} |\Gamma_\delta| \max_{(\tilde{f}, \tilde{g}) \in \Gamma_\delta} \mathbb{P} \left\{ \frac{1}{N} \left| \sum_i U_i \tilde{k}_i + \sum_j U_j \tilde{p}_j \right| + \delta \geq \frac{\epsilon}{4} (\alpha + \beta) \right. \\ & \quad \left. + \frac{\epsilon}{64(\lambda+1)B^2} \frac{1-\epsilon}{1+\epsilon} \left(\frac{1}{N} \left(\sum_i \tilde{k}_i^2 + \sum_j \tilde{p}_j^2 \right) - 8B^2(\lambda+1)\delta - \frac{\epsilon}{1-\epsilon} (\alpha + \beta) \right) \right\} \\ & \stackrel{(\dagger)}{\leq} |\Gamma_{\frac{\epsilon\beta}{5}}| \max_{(\tilde{f}, \tilde{g}) \in \Gamma_{\frac{\epsilon\beta}{5}}} \mathbb{P} \left\{ \frac{1}{N} \left| \sum_i U_i \tilde{k}_i + \sum_j U_j \tilde{p}_j \right| \geq \frac{\epsilon}{4} \alpha + \frac{\epsilon}{64(\lambda+1)B^2} \frac{1-\epsilon}{1+\epsilon} \left(\frac{1}{N} \left(\sum_i \tilde{k}_i^2 + \sum_j \tilde{p}_j^2 \right) - \frac{\epsilon}{1-\epsilon} \alpha \right) \right\}, \end{aligned}$$

where $(*)$ follows by observing that

$$\begin{aligned} & \frac{1}{N} \left| \sum_i U_i k_i + \sum_j U_j p_j \right| - \frac{1}{N} \left| \sum_i U_i \tilde{k}_i + \sum_j U_j \tilde{p}_j \right| \leq \frac{1}{N} \left| \sum_i U_i (k_i - \tilde{k}_i) + \sum_j U_j (p_j - \tilde{p}_j) \right| \\ & \leq \frac{1}{N} \left(\sum_i |k_i(f, g) - k_i(\tilde{f}, \tilde{g})| + \sum_j |p_j(f, g) - p_j(\tilde{f}, \tilde{g})| \right) \leq \delta, \end{aligned}$$

and

$$\begin{aligned} & \frac{1}{N} \left(\sum_i \tilde{k}_i^2 + \sum_j \tilde{p}_j^2 \right) - \frac{1}{N} \left(\sum_i k_i^2 + \sum_j p_j^2 \right) \leq \frac{1}{N} \left(\sum_i (\tilde{k}_i - k_i)(\tilde{k}_i + k_i) + \sum_j (\tilde{p}_j - p_j)(\tilde{p}_j + p_j) \right) \\ & \leq 8B^2(\lambda+1) \left(\sum_i |k_i(f, g) - k_i(\tilde{f}, \tilde{g})| + \sum_j |p_j(f, g) - p_j(\tilde{f}, \tilde{g})| \right) \leq 8B^2(\lambda+1)\delta. \end{aligned}$$

$(\dagger)$ follows by setting $\delta = \frac{\epsilon\beta}{5}$, and using

$$\begin{aligned} & \frac{\epsilon\beta}{4} - \frac{\epsilon\beta}{5} - \frac{\epsilon}{64(\lambda+1)B^2} \frac{1-\epsilon}{1+\epsilon} \left(8B^2(\lambda+1) \frac{\epsilon\beta}{5} + \frac{\epsilon}{1-\epsilon} \beta \right) = \frac{\epsilon\beta}{20} - \frac{\epsilon^2}{40} \frac{1-\epsilon}{1+\epsilon} \beta - \frac{\epsilon^2}{64(\lambda+1)B^2(1+\epsilon)} \beta \\ & \geq \frac{\epsilon\beta}{20} - \frac{\epsilon^2}{40} \frac{1-\epsilon}{1+\epsilon} \beta - \frac{\epsilon^2}{64B^2(1+\epsilon)} \beta \geq 0. \end{aligned}$$

Step 5: Bernstein's Inequality: Next we will show that by applying Bernstein's inequality, we get

$$\mathbb{P} \left\{ \frac{1}{N} \left| \sum_i U_i \tilde{k}_i + \sum_j U_j \tilde{p}_j \right| \geq \frac{\epsilon}{4} \alpha + \frac{\epsilon}{64(\lambda+1)B^2} \frac{1-\epsilon}{1+\epsilon} \left(\frac{1}{N} \left(\sum_i \tilde{k}_i^2 + \sum_j \tilde{p}_j^2 \right) - \frac{\epsilon}{1-\epsilon} \alpha \right) \right\} \leq 2 \exp \left(-\frac{N\epsilon^2(1-\epsilon)\alpha}{512(\lambda+1)B^2} \right). \quad (20)$$

For a fixed $(\tilde{f}, \tilde{g}) \in \Gamma_{\frac{\epsilon\beta}{5}}$, we define

$$\sigma^2 := \frac{1}{N} \left(\sum_i \tilde{k}_i^2 + \sum_j \tilde{p}_j^2 \right) = \frac{1}{N} \text{Var} \left(\sum_i U_i \tilde{k}_i + \sum_j U_j \tilde{p}_j \right),$$

$$A_1 := \frac{\epsilon}{4} \alpha - \frac{\epsilon^2}{64(\lambda+1)B^2(1+\epsilon)} \alpha, \text{ and } A_2 := \frac{\epsilon}{64(\lambda+1)B^2} \frac{1-\epsilon}{1+\epsilon}.$$

Then, by Bernstein's inequality, using the independence of the samples, we have

$$\begin{aligned} & \mathbb{P} \left\{ \frac{1}{N} \left| \sum_i U_i \tilde{k}_i + \sum_j U_j \tilde{p}_j \right| \geq \frac{\epsilon}{4} \alpha + \frac{\epsilon}{64(\lambda+1)B^2} \frac{1-\epsilon}{1+\epsilon} \left(\frac{1}{N} \left(\sum_i \tilde{k}_i^2 + \sum_j \tilde{p}_j^2 \right) - \frac{\epsilon}{1-\epsilon} \alpha \right) \right\} \\ &= \mathbb{P} \left\{ \frac{1}{N} \left| \sum_i U_i \tilde{k}_i + \sum_j U_j \tilde{p}_j \right| \geq A_1 + A_2 \sigma^2 \right\} \\ &\leq 2 \exp \left(-\frac{N(A_1 + A_2 \sigma^2)^2}{2\sigma^2 + \frac{8}{3}B^2(\lambda+1)(A_1 + A_2 \sigma^2)} \right) = 2 \exp \left(-\frac{3NA_2}{8B^2(\lambda+1)} \cdot \frac{(A_1/A_2 + \sigma^2)^2}{\left(\frac{3}{4B^2A_2(\lambda+1)} + 1 \right) \sigma^2 + \frac{A_1}{A_2}} \right) \\ &\stackrel{(\dagger)}{\leq} 2 \exp \left(-\frac{3NA_2}{8B^2(\lambda+1)} \cdot \frac{192\epsilon(1+\epsilon)(16(\lambda+1)B^2(1+\epsilon) - \epsilon)\alpha}{(\epsilon(1-\epsilon) + 48(1+\epsilon))^2} \right) \\ &\leq 2 \exp \left(-\frac{3NA_2}{8B^2(\lambda+1)} \cdot \frac{192\epsilon(1+\epsilon)16(\lambda+1)B^2\alpha}{(\epsilon(1-\epsilon) + 48(1+\epsilon))^2} \right) \\ &= 2 \exp \left(-\frac{3N}{8B^2(\lambda+1)} \cdot \frac{\epsilon(1-\epsilon)}{64(\lambda+1)B^2(1+\epsilon)} \cdot \frac{192\epsilon(1+\epsilon)16(\lambda+1)B^2\alpha}{(\epsilon(1-\epsilon) + 48(1+\epsilon))^2} \right) \\ &= 2 \exp \left(-\frac{18N\epsilon^2(1-\epsilon)\alpha}{(\lambda+1)B^2(\epsilon(1-\epsilon) + 48(1+\epsilon))^2} \right) \leq 2 \exp \left(-\frac{N\epsilon^2(1-\epsilon)\alpha}{512(\lambda+1)B^2} \right), \end{aligned}$$

where the last inequality follows using $\max_{0 \leq \epsilon \leq 1} (\epsilon(1-\epsilon) + 48(1+\epsilon))^2 = 9216$. $(\dagger)$ follows by noting that for any $u > 0$, $b > 2$, we have

$$\frac{(a+u)^2}{a+bu} \geq \frac{(a + \frac{b-2}{b}a)^2}{a + b \frac{b-2}{b}a} = 4a \frac{b-1}{b^2},$$

which yields that,

$$\begin{aligned} \frac{(A_1/A_2 + \sigma^2)^2}{\left(\frac{3}{4B^2A_2(\lambda+1)} + 1 \right) \sigma^2 + \frac{A_1}{A_2}} &\geq \frac{\frac{192A_1(1+\epsilon)}{\epsilon(1-\epsilon)}}{A_2 \left(1 + \frac{48(1+\epsilon)}{\epsilon(1-\epsilon)} \right)^2} = \frac{192\alpha\epsilon(1+\epsilon)(16(\lambda+1)B^2(1+\epsilon) - \epsilon)}{(\epsilon(1-\epsilon) + 48(1+\epsilon))^2} \\ &\geq \frac{192\epsilon(1+\epsilon)16(\lambda+1)B^2\alpha}{(\epsilon(1-\epsilon) + 48(1+\epsilon))^2}. \end{aligned}$$

Therefore (20) holds.

Step 6: Final Bound: Observe that if we have

$$\begin{aligned} \frac{1}{N} \left(\sum_i |f(X_i) - \tilde{f}(X_i)| + \sum_j |f(X_j) - \tilde{f}(X_j)| \right) &\leq \frac{\epsilon\beta}{40B}, \\ \frac{1}{N} \left(\sum_i |g(X_i) - \tilde{g}(X_i)| + \sum_j |g(X_j) - \tilde{g}(X_j)| \right) &\leq \frac{\epsilon\beta}{40B(\lambda+1)}, \end{aligned}$$

then we have

$$\begin{aligned} &\frac{1}{N} \left(\sum_i |k_i(f, g) - k_i(\tilde{f}, \tilde{g})| + \sum_j |p_j(f, g) - p_j(\tilde{f}, \tilde{g})| \right) \\ &\leq \frac{1}{N} \left(\sum_i 4B |f(X_i) - \tilde{f}(X_i)| + 4\lambda B |g(X_i) - \tilde{g}(X_i)| + \sum_j 4B |f(X_j) - \tilde{f}(X_j)| + 4B |g(X_j) - \tilde{g}(X_j)| \right) \leq \frac{\epsilon\beta}{5}, \end{aligned}$$

which implies that $\left| \Gamma_{\frac{\epsilon\beta}{5}} \right| \leq \mathcal{N}_1 \left(\mathcal{F}, \frac{\epsilon\beta}{40B}, X^N \right) \cdot \mathcal{N}_1 \left(\mathcal{G}, \frac{\epsilon\beta}{40(\lambda+1)B}, Z^N \right)$.

Thus, by combining all the steps together, we have the following.

$$\begin{aligned} &\mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \mathbb{E}[A_{f,g}] - \frac{1}{N} \sum_i k_i - \frac{1}{N} \sum_j p_j \geq \epsilon(\alpha + \beta + \mathbb{E}[A_{f,g}]) \right\} \\ &\leq \frac{16}{7} \mathcal{M}_1 \exp \left(-\frac{N\epsilon^2(1-\epsilon)\alpha}{512(\lambda+1)B^2} \right) + \frac{64}{7} \mathcal{M}_2 \exp \left(-\frac{\epsilon^2(1-\epsilon)(\alpha+\beta)N}{480B^4(\lambda+1)^2} \right) \end{aligned}$$

where $\mathcal{M}_1 = \max_{Z^N} \mathcal{N}_1 \left(\frac{\epsilon\beta}{40B}, \mathcal{F}, X^N \right) \cdot \mathcal{N}_1 \left(\frac{\epsilon\beta}{40(\lambda+1)B}, \mathcal{G}, Z^N \right)$, and

$$\begin{aligned} \mathcal{M}_2 &= \max_{Z^N} \left(\mathcal{N}_1 \left(\frac{\alpha+\beta}{10a}, \mathcal{G}_k, Z^n \right) + \mathcal{N}_1 \left(\frac{\alpha+\beta}{10b}, \mathcal{G}_p, Z^m \right) \right) \\ &\leq \max_{Z^n} \mathcal{N}_1 \left(\frac{\alpha+\beta}{80Ba}, \mathcal{F}, X^n \right) \cdot \mathcal{N}_1 \left(\frac{\alpha+\beta}{80Ba(\lambda+1)}, \mathcal{G}, Z^n \right) + \max_{Z^m} \mathcal{N}_1 \left(\frac{\alpha+\beta}{80Bb}, \mathcal{F}, X^m \right) \cdot \mathcal{N}_1 \left(\frac{\alpha+\beta}{80Bb}, \mathcal{G}, Z^m \right). \end{aligned}$$

Thus,

$$\begin{aligned} &\mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \mathbb{E}[A_{f,g}] - \frac{1}{N} \sum_i k_i - \frac{1}{N} \sum_j p_j \geq \epsilon(\alpha + \beta + \mathbb{E}[A_{f,g}]) \right\} \\ &\leq \frac{16}{7} \max_{Z^N} \mathcal{N}_1 \left(\frac{\epsilon\beta}{40B}, \mathcal{F}, X^N \right) \cdot \mathcal{N}_1 \left(\frac{\epsilon\beta}{40(\lambda+1)B}, \mathcal{G}, Z^N \right) \cdot \exp \left(-\frac{N\epsilon^2(1-\epsilon)\alpha}{512(\lambda+1)B^2} \right) \\ &\quad + \frac{64}{7} \max_{Z^n} \mathcal{N}_1 \left(\frac{\alpha+\beta}{80Ba}, \mathcal{F}, X^n \right) \cdot \mathcal{N}_1 \left(\frac{\alpha+\beta}{80Ba(\lambda+1)}, \mathcal{G}, Z^n \right) \cdot \exp \left(-\frac{\epsilon^2(1-\epsilon)(\alpha+\beta)N}{480B^4(\lambda+1)^2} \right) \\ &\quad + \frac{64}{7} \max_{Z^m} \mathcal{N}_1 \left(\frac{\alpha+\beta}{80Bb}, \mathcal{F}, X^m \right) \cdot \mathcal{N}_1 \left(\frac{\alpha+\beta}{80Bb}, \mathcal{G}, Z^m \right) \cdot \exp \left(-\frac{\epsilon^2(1-\epsilon)(\alpha+\beta)N}{480B^4(\lambda+1)^2} \right). \end{aligned}$$

Finally, we use scaling to get a cleaner bound for $\mu > 0$, by replacing B^2, α, β by $\mu B^2, \mu\alpha, \mu\beta$ as follows

$$\begin{aligned} &\mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \mathbb{E}[A_{f,g}] - \frac{1}{N} \sum_i k_i - \frac{1}{N} \sum_j p_j \geq \epsilon(\alpha + \beta + \mathbb{E}[A_{f,g}]) \right\} \\ &= \mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \mathbb{E}[a(\mu k) + b(\mu p)] - \frac{1}{N} \sum_i \mu k_i - \frac{1}{N} \sum_j \mu p_j \geq \epsilon(\mu\alpha + \mu\beta + \mathbb{E}[a(\mu k) + b(\mu p)]) \right\} \end{aligned}$$

$$\begin{aligned}
 &\leq \frac{16}{7} \left[\max_{Z^N} \mathcal{N}_1 \left(\frac{\epsilon\beta\sqrt{\mu}}{40B}, \mathcal{F}, X^N \right) \cdot \mathcal{N}_1 \left(\frac{\epsilon\beta\sqrt{\mu}}{40(\lambda+1)B}, \mathcal{G}, Z^N \right) \right] \exp \left(-\frac{N\epsilon^2(1-\epsilon)\alpha}{512(\lambda+1)B^2} \right) \\
 &+ \frac{64}{7} \left[\max_{Z^n} \mathcal{N}_1 \left(\frac{(\alpha+\beta)\sqrt{\mu}}{80Ba}, \mathcal{F}, X^n \right) \cdot \mathcal{N}_1 \left(\frac{(\alpha+\beta)\sqrt{\mu}}{80Ba(\lambda+1)}, \mathcal{G}, Z^n \right) \right. \\
 &\left. + \max_{Z^m} \mathcal{N}_1 \left(\frac{(\alpha+\beta)\sqrt{\mu}}{80Bb}, \mathcal{F}, X^m \right) \cdot \mathcal{N}_1 \left(\frac{(\alpha+\beta)\sqrt{\mu}}{80Bb}, \mathcal{G}, Z^m \right) \right] \cdot \exp \left(-\frac{\epsilon^2(1-\epsilon)(\alpha+\beta)N}{480B^4\mu(\lambda+1)^2} \right).
 \end{aligned}$$

Then by setting $\mu = \frac{1}{(\lambda+1)B^2}$, we have

$$\begin{aligned}
 &\mathbb{P} \left\{ \exists (f, g) \in \mathcal{F} \times \mathcal{G} : \mathbb{E}[A_{f,g}] - \frac{1}{N} \sum_i k_i - \frac{1}{N} \sum_j p_j \geq \epsilon(\alpha + \beta + \mathbb{E}[A_{f,g}]) \right\} \\
 &\leq \frac{16}{7} \left[\max_{Z^N} \mathcal{N}_1 \left(\frac{\epsilon\beta}{40B\sqrt{\lambda+1}}, \mathcal{F}, X^N \right) \cdot \mathcal{N}_1 \left(\frac{\epsilon\beta}{40(\lambda+1)^{3/2}B}, \mathcal{G}, Z^N \right) \right] \exp \left(-\frac{N\epsilon^2(1-\epsilon)\alpha}{512(\lambda+1)B^2} \right) \\
 &+ \frac{64}{7} \left[\max_{Z^n} \mathcal{N}_1 \left(\frac{(\alpha+\beta)}{80B^2\sqrt{\lambda+1}a}, \mathcal{F}, X^n \right) \cdot \mathcal{N}_1 \left(\frac{(\alpha+\beta)}{80B^2a(\lambda+1)^{3/2}}, \mathcal{G}, Z^n \right) \right. \\
 &\left. + \max_{Z^m} \mathcal{N}_1 \left(\frac{(\alpha+\beta)}{80B^2\sqrt{\lambda+1}b}, \mathcal{F}, X^m \right) \cdot \mathcal{N}_1 \left(\frac{(\alpha+\beta)}{80B^2\sqrt{\lambda+1}b}, \mathcal{G}, Z^m \right) \right] \cdot \exp \left(-\frac{\epsilon^2(1-\epsilon)(\alpha+\beta)N}{480B^2(\lambda+1)} \right) \\
 &\leq 10\mathfrak{M}_{\mathcal{F},\mathcal{G}}^{n,m,\lambda} \exp \left(-\frac{\epsilon^2(1-\epsilon)\alpha N}{512B^2(\lambda+1)} \right).
 \end{aligned}$$

Thus, we finished the proof of Theorem C.2. $\square$

C.3. Proof of Theorem 2.6

In Theorem C.2, we set $\epsilon = \frac{1}{2}$, $\alpha = \beta = \frac{t}{2}$, $C = \frac{1}{7680B^2}$, and we abbreviate $k_i(\hat{f}, \hat{g})$ as $\hat{k}_i$, $p_j(\hat{f}, \hat{g})$ as $\hat{p}_j$. Then

$$\begin{aligned}
 &\mathbb{P} \left\{ \mathbb{E}[\|(Y, Y) - (\hat{f}(X), \hat{g}(Z))\|_*^2] - \mathbb{E}[\|(Y, Y) - (f^*(X), g^*(Z))\|_*^2] - \frac{1}{N} \left(\sum_i \hat{k}_i + \sum_j \hat{p}_j \right) \right. \\
 &\quad \left. \geq \epsilon \left(\alpha + \beta + \mathbb{E}[\|(Y, Y) - (\hat{f}(X), \hat{g}(Z))\|_*^2] - \mathbb{E}[\|(Y, Y) - (f^*(X), g^*(Z))\|_*^2] \right) \right\} \\
 &\leq 5 \left(\max_{Z^N} \mathcal{N}_1 \left(\frac{t}{160B}, \mathcal{F}, X^N \right) \mathcal{N}_1 \left(\frac{t}{160(\lambda+1)B}, \mathcal{G}, Z^N \right) + \max_{Z^n} \mathcal{N}_1 \left(\frac{t}{80Ba}, \mathcal{F}, X^n \right) \mathcal{N}_1 \left(\frac{t}{80Ba(\lambda+1)}, \mathcal{G}, Z^n \right) \right. \\
 &\quad \left. + \max_{Z^m} \mathcal{N}_1 \left(\frac{t}{80Bb}, \mathcal{F}, X^m \right) \mathcal{N}_1 \left(\frac{t}{80Bb}, \mathcal{G}, Z^m \right) \right) \exp \left(-\frac{CNt}{\lambda+1} \right)
 \end{aligned}$$

Hence,

$$\begin{aligned}
 &\mathbb{E}_D \left[\mathbb{E}[\|(Y, Y) - (\hat{f}(X), \hat{g}(Z))\|_*^2] - \mathbb{E}[\|(Y, Y) - (f^*(X), g^*(Z))\|_*^2] - \frac{2}{N} \left(\sum_i \hat{k}_i + \sum_j \hat{p}_j \right) \right] \\
 &= \int_0^\infty \mathbb{P} \left\{ \mathbb{E}[\|(Y, Y) - (\hat{f}(X), \hat{g}(Z))\|_*^2] - \mathbb{E}[\|(Y, Y) - (f^*(X), g^*(Z))\|_*^2] - \frac{2}{N} \left(\sum_i \hat{k}_i + \sum_j \hat{p}_j \right) \geq t \right\} dt \\
 &\leq \delta + \int_\delta^\infty \mathbb{P} \left\{ \mathbb{E}[\|(Y, Y) - (\hat{f}(X), \hat{g}(Z))\|_*^2] - \mathbb{E}[\|(Y, Y) - (f^*(X), g^*(Z))\|_*^2] - \frac{2}{N} \left(\sum_i \hat{k}_i + \sum_j \hat{p}_j \right) \geq t \right\} dt
 \end{aligned}$$

$$\begin{aligned}
 &\leq \delta + \int_{\delta}^{\infty} 5 \left(\max_{Z^n} \mathcal{N}_1 \left(\frac{t}{80B\sqrt{\lambda+1}}, \mathcal{F}, X^n \right) \mathcal{N}_1 \left(\frac{t}{80B(\lambda+1)^{3/2}}, \mathcal{G}, Z^n \right) \right. \\
 &\quad + \max_{Z^m} \mathcal{N}_1 \left(\frac{t}{80B\sqrt{\lambda+1}}, \mathcal{F}, X^m \right) \mathcal{N}_1 \left(\frac{t}{80B\sqrt{\lambda+1}}, \mathcal{G}, Z^m \right) \\
 &\quad \left. + \max_{Z^N} \mathcal{N}_1 \left(\frac{t}{160B\sqrt{\lambda+1}}, \mathcal{F}, X^N \right) \mathcal{N}_1 \left(\frac{t}{160(\lambda+1)^{3/2}B}, \mathcal{G}, Z^N \right) \right) \exp \left(-\frac{CNt}{\lambda+1} \right) dt \\
 &\leq \delta + \int_{\delta}^{\infty} 5 \left(\max_{Z^n} \mathcal{N}_1 \left(\frac{1}{80B\sqrt{\lambda+1}n}, \mathcal{F}, X^n \right) \mathcal{N}_1 \left(\frac{1}{80Bn(\lambda+1)^{3/2}}, \mathcal{G}, Z^n \right) \right. \\
 &\quad + \max_{Z^m} \mathcal{N}_1 \left(\frac{1}{80B\sqrt{\lambda+1}m}, \mathcal{F}, X^m \right) \mathcal{N}_1 \left(\frac{1}{80B\sqrt{\lambda+1}m}, \mathcal{G}, Z^m \right) \\
 &\quad \left. + \max_{Z^N} \mathcal{N}_1 \left(\frac{1}{160B(m+n)\sqrt{\lambda+1}}, \mathcal{F}, X^N \right) \mathcal{N}_1 \left(\frac{1}{160(\lambda+1)^{3/2}B(m+n)}, \mathcal{G}, Z^N \right) \right) \exp \left(-\frac{CNt}{\lambda+1} \right) dt \\
 &= \delta + \exp(\mathfrak{R}_{\mathcal{F}', \mathcal{G}'}^{n,m,\lambda}) \cdot \frac{5(\lambda+1)}{CN} \exp \left(-\frac{CN\delta}{\lambda+1} \right) = \frac{(\lambda+1)(5 + \mathfrak{R}_{\mathcal{F}', \mathcal{G}'}^{n,m,\lambda})}{CN} \left(\text{Take } \delta = \frac{(\lambda+1)\mathfrak{R}_{\mathcal{F}', \mathcal{G}'}^{n,m,\lambda}}{CN} \geq \frac{1}{N} \right) \\
 &\leq \frac{6(\lambda+1)\mathfrak{R}_{\mathcal{F}', \mathcal{G}'}^{n,m,\lambda}}{CN},
 \end{aligned}$$

with

$$\mathfrak{R}_{\mathcal{F}', \mathcal{G}'}^{n,m,\lambda} := \log(T_n + T_m + T_{n+m}),$$

where for $k \in \{n, m, n+m\}$,

$$T_k := \sup_{Z^k} \mathcal{N}_1 \left(\frac{1}{c_k}, \mathcal{F}', X^k \right) \mathcal{N}_1 \left(\frac{1}{d_k}, \mathcal{G}', Z^k \right),$$

with

$$c_k = \begin{cases} 80B\sqrt{\lambda+1}k, & k \in \{n, m\}, \\ 160B\sqrt{\lambda+1}(n+m), & k = n+m, \end{cases}$$

and

$$d_k = \begin{cases} 80B(\lambda+1)^{3/2}k, & k = n, \\ 80B\sqrt{\lambda+1}k, & k = m, \\ 160B(\lambda+1)^{3/2}(n+m), & k = n+m. \end{cases}$$

Then observe that

$$\begin{aligned}
 \mathfrak{R}_{\mathcal{F}, \mathcal{G}}^{n,m,\lambda} &\leq \sup_{Z^N} \log \left(3 \mathcal{N}_1 \left(\frac{1}{d_N}, \mathcal{F}, X^N \right) \mathcal{N}_1 \left(\frac{1}{d_N}, \mathcal{G}, Z^N \right) \right) \leq \sup_{Z^N} \log \left(3 \left(\mathcal{N}_{1,\infty} \left(\frac{1}{d_N}, \mathcal{F} \times \mathcal{G}, Z^N \right) \right)^2 \right), \\
 &\lesssim \sup_{Z^N} \log \left(\mathcal{N}_{1,\infty} \left(\frac{1}{d_N}, \mathcal{F} \times \mathcal{G}, Z^N \right) \right),
 \end{aligned}$$

with $d_N = 160B(\lambda+1)^{3/2}N$.

Thus,

$$\mathbb{E}[\mathcal{L}(\hat{f}, \hat{g}; \lambda) - \mathcal{L}(f^*, g^*; \lambda)] - 2\mathbb{E}[\hat{\mathcal{L}}(\hat{f}, \hat{g}; \lambda) - \hat{\mathcal{L}}(f^*, g^*; \lambda)] \leq \frac{C_1(\lambda+1)\mathfrak{E}_{\mathcal{F} \times \mathcal{G}}^{N,\lambda}}{N},$$

for a universal constant $C_1 > 0$.

Hence, using that $\hat{\mathcal{L}}(\hat{f}, \hat{g}; \lambda) = \hat{\mathcal{L}}(f^*, g^*; \lambda) + \hat{\Delta}(\hat{f}, \hat{g})$, we obtain

$$\mathbb{E}_{\mathcal{D}}[\mathcal{L}(\hat{f}, \hat{g}; \lambda)] - \mathcal{L}(f^*, g^*; \lambda) \leq \frac{C_1(\lambda + 1)\mathfrak{E}_{\mathcal{F} \times \mathcal{G}}^{N, \lambda}}{m + n} + 2\mathbb{E}_{\mathcal{D}}[\hat{\Delta}(\hat{f}, \hat{g})]. \quad (21)$$

Combining Corollary 2.3 and (21), we finished our proof.

C.4. Proof of Remark 2.4 (ii)

Note that for any square-integrable random functions k_1 and k_2 , we have

$$1 - \frac{\mathbb{E}[\langle k_1, k_2 \rangle]^2]}{\mathbb{E}[\|k_1\|^2]\mathbb{E}[\|k_2\|^2]} = \frac{\mathbb{E}[\|k_1 - ck_2\|^2]}{\mathbb{E}[\|k_1\|^2]},$$

where

$$c = \frac{\mathbb{E}[\langle k_1, k_2 \rangle]}{\mathbb{E}[\|k_2\|^2]}.$$

Applying this identity with $k_1 = g^*(Z) - \hat{g}(Z)$ and $k_2 = f^*(X) - \hat{f}(X)$, we obtain

$$\begin{aligned} 1 - \rho_{\star}^2 &= \frac{\mathbb{E}_{\mathcal{D}, Z} \left[(g^*(Z) - \hat{g}(Z) - c(f^*(X) - \hat{f}(X)))^2 \right]}{\mathbb{E}_{\mathcal{D}, Z} \left[(g^*(Z) - \hat{g}(Z))^2 \right]} \\ &\geq \frac{\mathbb{E}_{\mathcal{D}, X} [\text{Var}(g^*(Z) - \hat{g}(Z) - c(f^*(X) - \hat{f}(X)) \mid \mathcal{D}, X)]}{\mathbb{E}_{\mathcal{D}, Z} \left[(g^*(Z) - \hat{g}(Z))^2 \right]} \\ &= \frac{\mathbb{E}_{\mathcal{D}, X} [\text{Var}(g^*(Z) - \hat{g}(Z) \mid \mathcal{D}, X)]}{\mathbb{E}_{\mathcal{D}, Z} \left[(g^*(Z) - \hat{g}(Z))^2 \right]}, \end{aligned}$$

where the inequality follows from the law of total variance.

Rearranging yields

$$\rho_{\star}^2 \leq \frac{\mathbb{E}_{\mathcal{D}, X} \left[\left(\mathbb{E}[g^*(Z) - \hat{g}(Z) \mid \mathcal{D}, X] \right)^2 \right]}{\mathbb{E}_{\mathcal{D}, Z} \left[(g^*(Z) - \hat{g}(Z))^2 \right]}.$$

C.5. Proof of Theorem 3.1

For comparison element $h = (f, g)$ with atomic norms:

$$C_f := \|f\|_{L^1(\mathcal{D}_f)}, \quad C_g := \|g\|_{L^1(\mathcal{D}_g)}, \quad C_h := C_f + C_g.$$

Define the excess energy:

$$a_k := \|r_k\|_{\star}^2 - \|(Y, Y) - h\|_{\star}^2$$

and the norms

$$\alpha_k = \|\Pi_{S_k^f} r_k\|_{\star}, \quad \beta_k = \|\Pi_{S_k^g} r_k^{(g)}\|_{\star}.$$

By orthogonal projection properties,

$$\|r_{k+1}\|_{\star}^2 = \|r_k^{(g)}\|_{\star}^2 - \beta_k^2 = \|r_k\|_{\star}^2 - \alpha_k^2 - \beta_k^2 = \|r_k\|_{\star}^2 - c_k^2,$$

where $c_k^2 = \alpha_k^2 + \beta_k^2 = a_k - a_{k+1}$. To prove Theorem 3.1, we first prove the following lemmas.

Lemma C.4. For every $k \geq 1$,

$$A_k = \sup_{a \in \mathcal{D}_f^{\star}} \langle r_k, a \rangle_{\star} \leq R_{\lambda} \alpha_k, \quad B_k = \sup_{b \in \mathcal{D}_g^{\star}} \langle r_k, b \rangle_{\star} \leq R_{\lambda} \alpha_k + R_{\lambda} \beta_k.$$

Proof. Write $r_k = r_k^{(g)} + \Pi_{S_k^f} r_k$ with $r_k^{(g)} \perp S_k^f$ by Step 1. For any $b \in \mathcal{D}_g^\star$,

$$\langle r_k, b \rangle_\star = \langle r_k^{(g)}, b \rangle_\star + \langle \Pi_{S_k^f} r_k, b \rangle_\star \leq \sup_{b \in \mathcal{D}_g^\star} \langle r_k^{(g)}, b \rangle_\star + R_\lambda \|\Pi_{S_k^f} r_k\|_\star,$$

where the last inequality uses Cauchy–Schwarz and $\|b\|_\star \leq R_\lambda$. For $a \in \mathcal{D}_f^\star$, by definition of $\psi_k^\star$ we have

$$\sup_{a \in \mathcal{D}_f^\star} \langle r_k, a \rangle_\star \leq R_\lambda \sup_{a \in \mathcal{D}_f^\star} \|\Pi_{S_{k-1}^f \cup \{a\}} r_k\|_\star = R_\lambda \|\Pi_{S_k^f} r_k\|_\star = R_\lambda \alpha_k.$$

Likewise,

$$\sup_{b \in \mathcal{D}_g^\star} \langle r_k^{(g)}, b \rangle_\star \leq R_\lambda \|\Pi_{S_k^g} r_k^{(g)}\|_\star = R_\lambda \beta_k,$$

hence $\sup_b \langle r_k, b \rangle_\star \leq R_\lambda \beta_k + R_\lambda \alpha_k$. This gives the second bound. $\square$

Lemma C.5. We have

$$\langle r_k, (Y, Y) \rangle_\star = \|r_k\|_\star^2 + \langle \Pi_{S_k^f} r_k, (f_{k-1}, 0) \rangle_\star, \quad (f_k, 0) = (f_{k-1}, 0) + \Pi_{S_k^f} r_k, \quad (22)$$

and, by polarization of the update,

$$\langle \Pi_{S_k^f} r_k, (f_{k-1}, 0) \rangle_\star = \frac{1}{2} \left(\|(f_k, 0)\|_\star^2 - \|(f_{k-1}, 0)\|_\star^2 - \|\Pi_{S_k^f} r_k\|_\star^2 \right). \quad (23)$$

Equivalently,

$$\langle r_k, (Y, Y) \rangle_\star = \|r_k\|_\star^2 + \frac{1}{2} \left(\|(f_k, 0)\|_\star^2 - \|(f_{k-1}, 0)\|_\star^2 - \|\Pi_{S_k^f} r_k\|_\star^2 \right). \quad (24)$$

Proof. By definition $r_k = (Y, Y) - (f_{k-1}, g_{k-1})$, and the Step 1 update is the orthogonal projection in the $\star$ -inner product, hence

$$(f_k, g_{k-1}) = (f_{k-1}, g_{k-1}) + \Pi_{S_k^f} r_k, \quad \text{so} \quad (f_k, 0) = (f_{k-1}, 0) + \Pi_{S_k^f} r_k.$$

For (22), write

$$\langle r_k, (Y, Y) \rangle_\star = \langle r_k, (f_{k-1}, g_{k-1}) + r_k \rangle_\star = \|r_k\|_\star^2 + \langle r_k, (f_{k-1}, g_{k-1}) \rangle_\star.$$

Since $(f_{k-1}, 0) \in \text{span}(S_k^f)$ and $\Pi_{S_k^f} r_k$ is the orthogonal projection onto S_k^f ,

$$\langle r_k, (f_{k-1}, 0) \rangle_\star = \langle \Pi_{S_k^f} r_k, (f_{k-1}, 0) \rangle_\star.$$

Moreover the previous step ends with $r_k \perp S_{k-1}^g$ and $(0, g_{k-1}) \in S_{k-1}^g$, hence $\langle r_k, (0, g_{k-1}) \rangle_\star = 0$, proving (22).

For (23), expand the square using $(f_k, 0) = (f_{k-1}, 0) + \Pi_{S_k^f} r_k$:

$$\|(f_k, 0)\|_\star^2 = \|(f_{k-1}, 0)\|_\star^2 + 2\langle \Pi_{S_k^f} r_k, (f_{k-1}, 0) \rangle_\star + \|\Pi_{S_k^f} r_k\|_\star^2,$$

and rearrange to obtain (23). Substituting (23) into (22) yields (24). $\square$

Lemma C.6. For every $k \geq 1$,

$$\sum_{i=2}^{k+1} a_i \leq 2\sqrt{2}C_h R_\lambda \sum_{i=1}^k c_i, \quad C_h = C_f + C_g, \quad c_i = \sqrt{a_i - a_{i+1}}.$$

Proof. Let $e := (Y, Y) - h$, $r_k := (Y, Y) - (f_{k-1}, g_{k-1})$, and recall

$$a_k := \|r_k\|_\star^2 - \|e\|_\star^2, \quad c_k^2 = a_k - a_{k+1} = \alpha_k^2 + \beta_k^2.$$

By Lemma C.5,

$$\|r_k\|_\star^2 = \langle r_k, (Y, Y) \rangle_\star - \langle \Pi_{S_k^f} r_k, (f_{k-1}, 0) \rangle_\star.$$

Then, by arithmetic–geometric mean inequality

$$\begin{aligned} a_k &= \|r_k\|_\star^2 - \|e\|_\star^2 = \langle r_k, e \rangle_\star + \langle r_k, (f, 0) \rangle_\star + \langle r_k, (0, g) \rangle_\star - \langle \Pi_{S_k^f} r_k, (f_{k-1}, 0) \rangle_\star - \|e\|_\star^2 \\ &\leq \frac{1}{2} a_k + \langle r_k, (f, 0) \rangle_\star - \langle \Pi_{S_k^f} r_k, (f_{k-1}, 0) \rangle_\star + \langle r_k, (0, g) \rangle_\star, \end{aligned}$$

and we use the bounds $\langle r_k, (f, 0) \rangle_\star \leq C_f A_k$ and $\langle r_k, (0, g) \rangle_\star \leq C_g B_k$ to control the two correlation terms, hence

$$\frac{a_k}{2} \leq C_f A_k + C_g B_k - \langle \Pi_{S_k^f} r_k, (f_{k-1}, 0) \rangle_\star.$$

Apply the polarization identity (Lemma C.5) to the f -history:

$$\langle \Pi_{S_k^f} r_k, (f_{k-1}, 0) \rangle_\star = \frac{1}{2} \left(\|(f_k, 0)\|_\star^2 - \|(f_{k-1}, 0)\|_\star^2 - \alpha_k^2 \right).$$

Therefore,

$$\frac{a_k}{2} \leq C_f A_k + C_g B_k + \frac{1}{2} \left(\|(f_{k-1}, 0)\|_\star^2 - \|(f_k, 0)\|_\star^2 \right) + \frac{1}{2} \alpha_k^2.$$

Summing over $i = 1, \dots, k$ (with $f_0 = 0$) yields

$$\sum_{i=1}^k \frac{a_i}{2} \leq C_f \sum_{i=1}^k A_i + C_g \sum_{i=1}^k B_i - \frac{1}{2} \|(f_k, 0)\|_\star^2 + \frac{1}{2} \sum_{i=1}^k \alpha_i^2. \quad (25)$$

Since $\sum_{i=1}^k \alpha_i^2 \leq \sum_{i=1}^k c_i^2 = \sum_{i=1}^k (a_i - a_{i+1}) = a_1 - a_{k+1}$, $A_i \leq R_\lambda \alpha_i$, and $B_i \leq R_\lambda \alpha_i + R_\lambda \beta_i \leq \sqrt{2} c_i$, we get

$$C_f \sum_{i=1}^k A_i + C_g \sum_{i=1}^k B_i \leq C_f R_\lambda \sum_{i=1}^k c_i + \sqrt{2} C_g R_\lambda \sum_{i=1}^k c_i \leq \sqrt{2} (C_f + C_g) R_\lambda \sum_{i=1}^k c_i = \sqrt{2} C_h R_\lambda \sum_{i=1}^k c_i.$$

Dropping the nonpositive term $-\frac{1}{2} \|(f_k, 0)\|_\star^2$ in (25) gives

$$\sum_{i=1}^k \frac{a_i}{2} \leq \sqrt{2} C_h R_\lambda \sum_{i=1}^k c_i + \frac{a_1 - a_{k+1}}{2},$$

which implies

$$\sum_{i=2}^{k+1} a_i \leq 2\sqrt{2} C_h R_\lambda \sum_{i=1}^k c_i.$$

□

We are ready to prove Theorem 3.1.

Proof. Set

$$S_k := \sum_{i=1}^k a_i, \quad H_k := \sum_{i=1}^k \frac{1}{i}.$$

We have that for every $k \geq 1$,

$$\sum_{i=2}^{k+1} a_i \leq 2\sqrt{2} C_h \sum_{i=1}^k c_i, \quad c_i = \sqrt{a_i - a_{i+1}}. \quad (26)$$

Summation by parts (discrete Abel transform): For every $k \geq 1$,

$$\sum_{i=1}^k i(a_i - a_{i+1}) = \sum_{i=1}^k i c_i^2 = S_k - k a_{k+1}. \quad (27)$$

Indeed,

$$\sum_{i=1}^k i(a_i - a_{i+1}) = \sum_{i=1}^k \sum_{j=1}^i (a_i - a_{i+1}) = \sum_{j=1}^k \sum_{i=j}^k (a_i - a_{i+1}) = \sum_{j=1}^k (a_j - a_{k+1}) = S_k - k a_{k+1}.$$

Cauchy–Schwarz with weights and (27): By Cauchy–Schwarz,

$$\left(\sum_{i=1}^k c_i \right)^2 = \left(\sum_{i=1}^k \frac{1}{\sqrt{i}} \sqrt{i} c_i \right)^2 \leq \left(\sum_{i=1}^k \frac{1}{i} \right) \left(\sum_{i=1}^k i c_i^2 \right) = H_k (S_k - k a_{k+1}) \leq H_k S_k.$$

Using (26) and $S_k \leq S_{k+1}$,

$$S_{k+1} - a_1 = \sum_{i=2}^{k+1} a_i \leq 2\sqrt{2} C_h R_\lambda \sum_{i=1}^k c_i \leq 2\sqrt{2} C_h R_\lambda \sqrt{H_k S_{k+1}}.$$

Then

$$S_{k+1} \leq a_1 + 2\sqrt{2} C_h R_\lambda \sqrt{H_k S_{k+1}} \leq a_1 + S_{k+1}/2 + 4C_h^2 R_\lambda^2 H_k,$$

so $S_{k+1} \leq 2a_1 + 8C_h^2 R_\lambda^2 H_k$. Hence for all $k \geq 1$,

$$S_k \leq 2a_1 + 8C_h^2 R_\lambda^2 H_k. \quad (28)$$

Pointwise rate: Since (a_k) is decreasing, $k a_k \leq S_k$. Combining with (28),

$$a_k \leq \frac{2a_1}{k} + \frac{8C_h^2 R_\lambda^2 H_k}{k} \leq C \frac{C_h^2 R_\lambda^2 \log k}{k},$$

for a universal constant $C > 0$, completing the proof. $\square$